

A (Pan-Canadian) Cluster Randomized Control Effectiveness Trial of the ABRACADABRA Web-Based Literacy Program

Robert Savage
McGill University

Philip C. Abrami
Concordia University

Noella Piquette
Lethbridge University

Eileen Wood
Wilfrid Laurier University

Gia Deleveaux
Concordia University

Sukhbinder Sanghera-Sidhu and Giovani Burgos
McGill University

This report describes a cluster randomized control trial (RCT) intervention study of the effectiveness of the ABRACADABRA (ABRA) Web-based literacy system using a classroom-level RCT intervention with 1,067 children in 74 kindergarten and Grade 1 or Grade 1/2 classrooms across Canada. The authors closely followed the CONSORT criteria for executing and reporting high-quality RCT studies. Well-trained teachers delivered the ABRA intervention to their regular classrooms for 20 hr per child over one full semester. At posttest, the ABRA intervention classroom showed significant advantages over controls in phonological blending ability, letter–sound knowledge and, marginally, for phoneme segmentation fluency. A secondary analysis exploring the effects of different levels of program implementation showed that with fidelity of implementation (80% of intervention teachers), advantages were evident at posttests in phonological blending, phoneme segmentation fluency, sight word reading, and letter–sound knowledge. It is concluded that ABRA is an effective resource for key skills associated with early reading. Implications for the role of both Web-based technologies and extended professional development for technology in aiding in the scale-up of evidence-based reading interventions are discussed.

Keywords: reading, intervention, technology, randomized control trial

What evidence is there that technology can aid reading acquisition? Up until very recently, it could be fairly remarked that much enthusiasm exists for the potential of computer-based information and communication technologies (ICT) for helping children in learning to read (Bereiter, 2002; Dede, 1996; Harasim, Hiltz, Teles, & Turoff, 1995; Mayer, 2001; Rabiner & Malone, 2004; Scardamalia & Bereiter, 1996). There is also widespread deployment of ICT in North American schools (Chambers,

Abrami, et al., 2008; Cuban, 2001). Nevertheless, expectation in this domain has run far in advance of the existing evidence base (Andrews et al., 2007; Chambers, Abrami, et al., 2008; Savage, Abrami, Hips, & Deault, 2009). Beyond the effectiveness of technology, there is also a need to closely tie ICT use to contemporary theoretical models of reading acquisition and to coherent pedagogical models for technology.

Turning first to the evidence base for technology and reading, several reviews of research that have included quasi-experimental studies have identified small effect sizes for ICT on literacy, and researchers are therefore cautiously optimistic about ICT (Blok, Oostdam, Otter, & Overmatt, 2002; Ehri et al., 2001; MacArthur, Ferretti, Okolo, & Cavalier, 2001). However, to confidently answer the question, “Does technology cause improvement in reading?” methodologically, high-quality randomized control trials (RCTs) must occupy a central role, as they provide unique protection that effects reported are due to the intervention. Finally, systematic reviews of the whole literature including *only* such studies provide the greatest confidence in findings (Savage, 2012; Torgerson, Brooks, & Hall, 2006).

In this specific methodological sense then, the most reliable evidence concerning ICT can be derived from three comprehensive systematic reviews (Andrews et al., 2007; Slavin, Cheung, Groff, & Lake, 2008; Slavin, Lake, Chambers, Cheung, & Davis, 2009; Torgerson & Zhu, 2003). Torgerson and Zhu (2003) found

This article was published Online First December 17, 2012.

Robert Savage, Department of Educational and Counseling Psychology, McGill University, Montreal, Quebec, Canada; Philip C. Abrami, Centre for the Study of Learning and Performance, Concordia University, Montreal, Quebec, Canada; Noella Piquette, Faculty of Education, Lethbridge University, Lethbridge, Alberta, Canada; Eileen Wood, Department of Developmental Psychology, Wilfrid Laurier University, Waterloo, Ontario, Canada; Gia Deleveaux, Centre for the Study of Learning and Performance, Concordia University; Sukhbinder Sanghera-Sidhu, Department of Educational and Counseling Psychology, McGill University; Giovani Burgos, Department of Sociology, McGill University.

Correspondence concerning this article should be addressed to Robert Savage, Department of Educational and Counseling Psychology, Faculty of Education, McGill University, Montreal H3A 1Y2, Canada. E-mail: robert.savage@mcgill.ca

only 12 RCT studies among an initial 2,319 studies that examined ICT-based interventions for children ages 5–16 years. Torgerson and Zhu found that across these 12 studies, some effect sizes were positive, and some, including those of the largest RCT study, were actually negative. Overall, only small and nonsignificant effect sizes for interventions were evident. Torgerson and Zhu (2003) concluded that teachers should not use technology to teach literacy until several, new, well-designed RCT studies with consistently positive effects have been published and evaluated (see also Torgerson, 2007).

Similarly, a systematic review of technology on middle and high school learning by Slavin et al. (2008) found few ($n = 8$) well-designed intervention studies. The mean overall effect size was a very modest $+0.10$, suggesting that computers do not aid literacy skill development in any substantial way. Slavin et al. (2009) also reported similar patterns for 12 elementary school studies. It is important to note that Slavin et al. reviewed only published articles and did not seek unpublished reports, which may inflate findings due to a publication bias. Their studies also included quasi-experiments in addition to RCT studies.

Similar trends are evident in much of the recent research literature. Using a large-scale RCT study design, Dynarski et al. (2007) evaluated technology-based reading interventions using a range of commercially available ICT products in 132 schools with a total of 4,389 teachers. Dynarski et al. found that the mean effect size for interventions was not significantly different from zero on any tests of reading in either Grade 1 or Grade 4. In a follow-up study, Campuzano, Dynarski, Agodini, Rall, and Pendeleton (2009) explored a representative subset of classrooms from the Dynarski et al. study and again reported almost no significant effects of technology intervention on attainment and no effects on literacy whatsoever.

Finally, Andrews et al. (2007) carried out a systematic review of the impact of technology on written language learning. They identified only eight reasonably well-designed intervention studies from an initial search of 2,103 recent articles and reports, of which only one of the included studies (Lewis, Ashton, Haapa, Kieley, & Fielden, 1999) had used some random allocation of participants. Andrews et al. concluded that while many studies purport to have demonstrated effectiveness of ICT, none have done so convincingly. Andrews et al. also drew attention to the lack of clear definition of terms and the absence of theorizing of the role of technology in supporting literacy. They summed up the state of the literature on ICT as follows:

[T]he field is in a preparadigmatic state where definitions of English, literacy, and ICT are still relatively unclear and where the causal or reciprocal relationships between them have yet to be fully theorized. (p. 325)

Methodological Issues in Interpreting Effects of ICT

In evaluating the pessimistic findings of this RCT-based literature, there are perhaps three overarching methodological issues that should be noted. The first issue concerns *implementation* of studies, the second issue concerns the *quality of the technology* used in research studies, and a final issue concerns what we term the *theoretical and pedagogical coherence* of technologies and their implementation. Turning first to implementation, teacher-led reading intervention research suggests that the quality of the im-

plementation of programs has a significant impact on outcomes both when the interventions involve technology (Chambers, Abrami, et al., 2008; Savage et al., 2010; Wolgemuth et al., 2011) and when they do not involve technology (e.g., Davidson, Fields, & Yang, 2009; Lane, Bocian, MacMillan, & Gresham, 2004; Stein et al., 2008). Indeed, in several recent studies, variation in program implementation was the biggest factor moderating outcomes and could explain *entirely* the implementation effects reported (e.g., Chambers, Abrami, et al., 2008; Davidson et al., 2009). Rigorous and detailed investigation of the ways teachers implement (or do not implement) interventions clearly needs to be explored in detail in ICT studies, yet this is rarely reported. In the Campuzano et al. (2009) study, for example, there was no assessment of treatment integrity whatsoever. This issue of teachers' fidelity to treatment protocols is particularly problematic in ICT-based interventions. Much research shows that a host of teacher factors—including teachers' use of ICT at home (Wozney et al., 2006); comfort with computers (Chen & Chang, 2006); beliefs and expertise (Chen, 2008; Sang, Valcke, van Braak & Tondeur, 2010), and fear and other emotional reactions toward computers—affect teachers' actual use of computer technology (see Kay, 2008; Mueller, Wood, & Willoughby, 2008; Wood, Specht, Willoughby, & Mueller, 2008).

Some formal models of teacher's use of technology posit stages of development (e.g., Mueller, Wood, Willoughby, Ross, & Specht, 2008; Sandholtz, Ringstaff, & Dwyer, 1997). In the teachers' use of technology model by Sandholtz et al., the *entry* level is characterized by the making of time-consuming mistakes, frustration, and high levels of discontinuation. Teachers at the *adoption* level can use technology in a systematic manner. Adoption is characterized adherence to treatment protocols (treatment integrity) with minimum experimentation or formation of links to other forms of learning, such as collaborative or experientially based learning. Adaptation, by contrast, steps beyond conventional treatment integrity. At the *adaptation* level, teachers' integration of technology has been described as transforming classroom teaching, with greater connectivity between all forms of learning. Savage et al. (2010) found that evidence of all three of these types of practice in the use of an Internet-based reading software program (ABRACADABRA, A Balanced Reading Approach for Canadians Designed to Achieve Best Results for All). Furthermore, teachers rated as being at the adaptation level produced significantly greater student gains in reading than teachers at other levels of practice. Wolgemuth, Helmer, Bottrell, Harper, and Lea (2012) reported comparable effects of implementation quality on student outcomes. In another model, Wozney, Venkatesh, and Abrami (2006) used expectancy theory to examine teachers' use of technology related to three broad motivational categories: perceived *expectancy* of success, perceived *value* of technology use, and perceived *cost* of technology use. Each category impacted on use. Expectancy beliefs mattered the most; value factors were positively associated with use, but cost factors were negatively associated with use.

A closely related issue concerns the training given for technology use in intervention research. Teachers are unlikely to successfully implement any program if they have limited training and support in its use in the classroom. It is nevertheless quite typical in ICT studies to offer a single day of training despite the widespread understanding that such models of professional learning

produce extremely limited professional change (e.g., Clarke & Hollingsworth, 2002; Fullan, 2001; Waks, 2007) and are widely discredited “straw man” models. The Dynarski et al. (2007) study was quite typical in this regard: Teachers were given a single-day training session on the technology several weeks before the intervention commenced. Teacher self-reports showed that around a third of teachers did not feel qualified to use the technology at the start of the intervention. As noted earlier, the observed quality of classroom delivery of ICT was not explored, so the true capacity of teachers to deliver the trained interventions is unknown. The quality of teacher training has also not been reported as a factor in systematic reviews of ICT effectiveness.

It is possible, however, to effectively support teachers in their use of new technology. Anderson, Wood, Piquette-Tomei, Savage, and Mueller (2011) documented all requests from 10 teachers in kindergarten and Grades 1 and 2 who received “just-in-time” instructional support over a 2-1/2-month period while implementing a novel technology-based intervention. From detailed observations, subsequent analysis of 80 just-in-time support sessions indicated that the greatest level of support was required during the initial stages of implementation and for technological rather than pedagogical questions. Support requests declined sharply over the first 3 weeks or so of implementation, and most of the problems were resolved immediately through the just-in-time support system. These results inform the kinds of ongoing supports teachers need to run technology-based interventions effectively.

The second issue is that whether ICT “works” will always be subject to the quality of ICT available. In many studies, experimenters rightly attempt to pick the best technologies available. Dynarski et al. (2007) selected ICT that, according to the manufacturers, had first been research validated. There was, however, no prior independent assessment of the nature and quality of the technology programs used. Recently, Grant et al. (2012) developed a taxonomy of cognitive reading skills based upon both systematic reviews of evidence (e.g., National Reading Panel, 2000) and an expert panel. Grant et al. then used the taxonomy to explore how well 30 commercially available software programs designed to teach early literacy support these key reading skills. Results showed that although some skills such as alphabetic knowledge were being trained in a somewhat developmentally appropriate manner, very few programs (15%) taught synthetic phonics, and none at all taught phoneme segmentation or concepts about print. Key comprehension skills, such as summarizing a text or generating questions, were entirely absent. Overall Grant et al. noted that there were limited examples for training each skill, inconsistent progression, and few opportunities to practice skills. The study by Grant et al. suggests that better evidence-based ICT is needed if ICTs are to be used to assist in reading skill development.

A related final issue that we would draw attention to in evaluating technology is the link between contemporary theories of reading developed inside and outside technology and the pedagogical theory of implementation of ICT. Among the best of recent work in this field are the reports by Chambers, Abrami, et al. (2008) and Chambers, Slavin, et al. (2008) on a series of well-designed RCT studies. Each study found significant positive effects of a technology on literacy used to complement the nontechnology-based Success for All program. Alongside this empirical evidence, Chambers, Slavin, et al. (2008) also explicitly identified a pedagogical role for technology: They articulated what

computers cannot do and delineated a role for the technology as adding value to regular classroom teaching in what they refer to as *embedded ICT*. In addition, Chambers et al. drew upon contemporary theory about how ICT supports reading through encouraging dual visual and verbal coding (Clark & Paivio, 1991) and through “offloading” between modalities to reduce working memory load, (Solso, 2001) thereby encouraging retention.

Beyond these examples, it is rare in ICT studies on literacy to consider contemporary cognitive models of reading. To take certain aspects of word recognition as an example, all current accounts of word reading assume strong connectivity between representations of words or sublexical letter strings (Bishop & Snowling, 2004; Coltheart, Curtis, Atkins, & Haller, 1993; Seidenberg & McClelland, 1989). A key issue thus concerns the specificity of spelling representations in these distributed lexicons. In dominant-phase-based models, partly specified phonologically based cues in words are developmentally important (e.g., Ehri, 2005; Perfetti, 2007; Savage & Stuart, 2006; Share, 1995). Phonological awareness is assumed to be the cutting edge of word learning (Hulme, Snowling, Caravolas, & Carroll, 2005; Share, 1995) and must be applied fluently to word recognition in text. This analysis suggests the need for progression within ICT activities that reflects these qualitative changes in development (see also Ecalte, Magnan, & Calmus, 2009, for theoretically driven research on ICT).

Naturally, there are also a number of unresolved issues. One example is the relative role of rhyme-based inference and use of grapheme–phoneme rules in early reading (see Savage, Deault, Daki, & Aouad, 2011, for a recent review). Another is the relative efficacy of different kinds of phonics programs such as “synthetic phonics” and “analytic phonics” (Torgerson et al., 2006). Wise implementation of ICT in this uncertainty would allow *all* candidate models to be further tested fairly; yet of the 30 ICT products reported by Grant et al. (2012), nearly all implemented only one model of phonics. For all of these reasons, we have over several years developed the ABRACADABRA reading intervention as a means to address previous short comings in literacy ICT programs.

The ABRACADABRA Web-Based Literacy Programmatic Research

As mentioned earlier, ABRACADABRA (hereafter, ABRA), is an acronym for A Balanced Reading Approach for Canadians Designed to Achieve Best Results for All. ABRA is a free-access Web-based literacy software for beginning readers. ABRA has four modules: letters and sounds, reading, comprehension, and writing/spelling. It can be used flexibly. ABRA does not need to be used within the context of a particular nontechnology reading program. It contains 32 distinct leveled activities and 17 stories. ABRA was developed by a multiuniversity team (see Abrami et al., 2010; Abrami, Savage, Wade, Hipps, & Lopez, 2008; Hipps, Abrami, Savage, Cerna, & Jorgensen, 2005, for details). ABRA is based on the evidence from systematic reviews on effective reading interventions as well as developmental theory (see Savage et al., 2009, for a review) and includes instructional, professional, and parent modules (see <http://abralite.concordia.ca>). A version of ABRA containing an assessment and communication module is also freely available; however, due to the fact that it can store

student records, this must be downloaded and stored on a school board server (see <http://grover.concordia.ca/abracadabra/>).

ABRA is supported by evidence from three RCT efficacy studies. In a within-classroom RCT design, Comaskey, Savage, and Abrami (2009) sampled 53 children from a disadvantaged low socioeconomic (SES) urban kindergarten where the majority of students experienced English as a second language. Children were exposed to ABRA instruction in small groups for 10 hr over 13 weeks. Phoneme-based teaching led to significant growth in phoneme blending in the ABRA group only. Rhyme-based teaching led to similarly specific growth in rhyme awareness. Di Stasio, Savage, and Abrami (2012) followed up these children 1 year later. They reported that the analytic phonics group performed better on a passage-reading comprehension task than the synthetic phonics group. An overall effect size of 0.41 per hour of intervention was reported.

Savage et al. (2009) also explored the effectiveness of ABRA in 144 typically developing Grade 1 children using a within-classroom RCT design and also contrasted phoneme-based “synthetic” phonics and a rhyme-based “analytic phonics” methods as well as a regular classroom control. Children received the intervention in small groups for an average of 13 hr per child delivered by trained research assistants. Significant differences between treatment and control groups at posttest were evident on phonological awareness, listening comprehension and reading fluency (favoring synthetic phonics), and letter-sound knowledge (favoring analytic phonics). In addition, overall effect sizes were significantly different from zero at immediate and delayed (8 months) posttest after the completed intervention phase in both interventions. In addition, Deault, Savage, and Abrami (2009) explored response characteristics of students in this sample and found that ABRA can moderate the associations between literacy and attention and may support students at risk of reading and attention difficulties.

Finally, Wolgemuth et al. (2012) reported a within-class RCT evaluating ABRA in the Northern Territories in Australia with over 300 children, including a large number of aboriginal students for whom English was an additional language. Specially recruited and trained teachers delivered ABRA as a pull-out program in schools for around 40 min four times a week for 16 weeks. At posttest, the children who received the ABRA intervention showed significant advantages in phonological awareness ($d = 0.37$) and grapheme-to-phoneme knowledge ($d = 0.37$). Aboriginal students made at least the same amount of progress as nonaboriginal students and, sometimes, greater progress.

The Current Study

All current scientific evidence regarding ABRA comes from *efficacy trials*—trials run by university-based researchers and assistants or delivered by specially trained professionals. However, ABRA was developed as a free tool for teachers and placed on the worldwide Web to allow maximal accessibility and scalability of interventions (e.g., Abrami et al., 2008). The next step necessary to establish the impact of this ICT software is large-scale *effectiveness trials* to explore the impact of interventions under real-world conditions undertaken by well-supported regular classroom teachers within their regular classrooms. In such approaches, classrooms, not students, are treated as the unit of analysis. Well-trained

teachers with the ability to implement ICT interventions thus have a central role in such designs. As there is much evidence of the impact of variation in teachers’ delivery of programs, a secondary question concerns the impact of adequate adoption of technology over inadequate adoption and of the possible role of adapted implementation of ABRA.

Thus, there are two main research questions of this article: (a) Does a classroom-level-cluster RCT effectiveness trial of ABRA yield significant advantages for intervention over control classrooms in early literacy at posttest? and (b) Do adequately implementing classrooms and transformational implementations in the classrooms yield additional effects for ABRA?

Method

Design

This study is a cluster RCT intervention study that took place over two academic years, 2007–2008 and 2008–2009. This study used a pretest–posttest experimental intervention design and randomized 74 participating classrooms containing 1,067 students) across the three provinces of Alberta, Ontario, and Quebec, Canada. Randomization took place within schools at the classroom level. Pairs of classrooms were identified within each school and were then randomly allocated to either ABRA intervention ($n = 37$ classrooms) or control non-ABRA regular classroom teaching conditions ($n = 37$ classrooms). Such prior identification reduces bias in cluster RCT designs (Puffer, Torgerson, & Watson, 2005). Randomization was achieved by blindly pulling slips of paper containing teachers’ names from a hat in each school.

We made a formal power calculation using Power and Precision Version 3, (Borenstein, Rothstein, Cohen, Schoenfeld, & Berlin, 2001). This showed that with α set at .05, $n = 35$ cases in each cell, and two covariates, this design would yield a .88 power to detect a large effect size in standard analysis of covariance (ANCOVA, where a large effect is defined, according to Cohen, 1988, as $f = 0.40$). This level of power appeared very reasonable given that Savage et al. (2009) reported large effect sizes for ABRA on standardized measures of phonological awareness.

Students in the control classrooms continued to receive their regular English language arts (ELA) lessons that did not include ABRA, while those in the experimental group had ABRA integrated into their ELA lessons. ABRA was part of regular classroom teaching and was not a supplementary program. Inclusion of ABRA did not result in extra pedagogical time being allocated to literacy in the intervention classrooms. Beyond the ABRA intervention, all teachers in both study conditions were encouraged to carry on ‘business as usual’ in their teaching. The conceptual and methodological issues behind this request were explained to teachers and often revisited. Intervention teachers implemented the ABRA Web-based literacy program 2 hr per week. Posttesting took place after 10–12 weeks of intervention.

Sample and Participant Selection

Classroom teacher recruitment. Starting in the spring of 2007, members of the research team made demonstrations of the ABRA software to principals, teachers, ELA consultants at school

boards, and other educators at various meetings throughout Quebec, Ontario, and Alberta. School administrators and school board officials who then expressed interest received letters asking for their participation. These letters outlined expectations for involvement. Principals of participating schools were required to allow random allocation of classrooms to the intervention or control condition, to ensure that the experimental classes would be guaranteed 2 hr of computer time each week for ABRA implementation, and to allow teachers assigned to the experimental condition to attend ABRA training sessions. In return, participating schools received free ABRA teacher technological and pedagogical training and support. Control classroom teachers received ABRA training after the intervention was completed.

Schools approached for this study were nonselective urban, suburban, and rural elementary schools. From the recruitment presentations, 81 teachers from 24 schools across the three participating provinces responded positively. However, from the group of 81 classrooms that participated in the study, seven of the classrooms involved did not meet the random assignment requirements to be part of the final RCT analysis. Field notes on the recruitment and selection process of the participating classrooms indicated that three of the classroom teachers were not willing to be randomly assigned. Data collected on a delayed intervention group from Ontario also were removed, as this classroom had not been matched to a control comparison group. Two additional intervention groups were removed: a Grade 1–2 split class from Ontario and a Grade 1 class from Quebec, as neither class had been matched with a control comparison classroom within their respective schools. Similarly, data from a kindergarten control class from Quebec were removed, as this class had not been matched with an intervention comparison group. The final ABRA RCT data set contained 74 matched classrooms, consisting of 12 paired kindergarten classrooms ($n = 24$) and 25 paired first- and second-grade classrooms ($n = 50$). The details of the final sample analyzed are specified in the Enrollment and Allocation sections of the CONSORT flow diagram (see Figure 1).

Student participants. Once classroom teachers had agreed to participate in the ABRA study, investigators obtained permission for their students to participate. There were 1,068 students whose parents agreed to allow their child(ren) to participate in the study. One consented kindergarten student moved away before pretesting, which brought the sample total to 1,067 students. The child sample included 316 kindergarten students ($n = 154$ ABRA, $n = 162$ control), 616 first graders ($n = 352$ ABRA, $n = 264$ control), and 135 second grade students ($n = 43$ ABRA, $n = 92$ control). By gender, the final sample of children consisted of 543 girls and 524 boys. The random allocation of classrooms resulted in 284 girls and 265 boys in the experimental group (total $n = 549$), and 259 girls and 259 boys in the control group (total $n = 518$). Age of the participating students at pretest averaged 73.69 months old ($SD = 10.11$ months), with a range of 67 months. No student was excluded due to language or exceptionalities.

The demographic profile of schools was evaluated through a formal report created the Canadian Institute for Social Policy (CRISP; D. Willms, personal communication, April 22, 2011). Here, individual student-level postal code data for all schools in the study ($n = 24$ schools) were linked to data from the long form of the 2006 Canadian Census (Statistics Canada, 2006) to construct a set of derived variables that describe the SES characteristics of

A CONSORT E-flowchart at the classroom level for the Pan-Canadian ABRA Cluster RCT study.

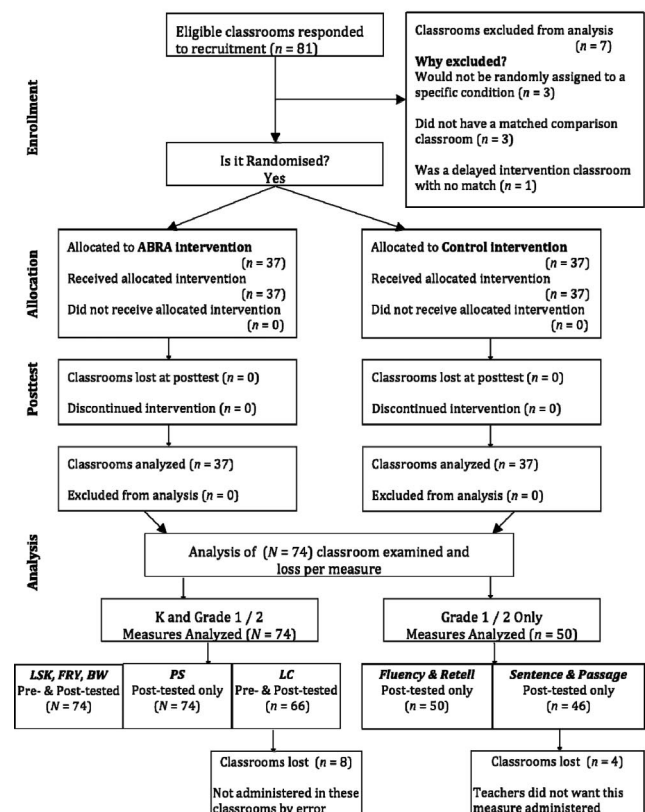


Figure 1. Consolidated Standards of Reporting Trials (CONSORT) E-flowchart showing classroom level for the cluster randomized clinical trial study of the Pan-Canadian ABRA (acronym for ABRACADABRA, A Balanced Reading Approach for Canadians Designed to Achieve Best Results for All). LSK = Letter-Sound Knowledge; FRY = Fry Words; BW = Blending Words subtest; PS = Phonemic Segmentation Fluency subtest; LC = Listening Comprehension measure.

the postal code area in which each child resides. A composite z -standardized measure was derived on the basis of occupational status, unemployment rates, family income, and years of education with national mean of 0 and standard deviation of 1. The study sample scores were normally distributed, with a mean of 0.03 ($SD = 0.55$). This confirmed that schools in our sample closely reflected national school SES patterns.

Procedure

ABRACADABRA sessions. Before intervention, teachers began implementing ABRA in their classrooms; a full day of training took place led by three of the research investigators. Teachers were exposed to the philosophical, developmental, and pedagogical underpinnings of the software and were given hands-on time to explore the software. Theoretically based developmental progression was emphasized throughout training. For example, teachers were shown how the ABRA phonic activities follow theoretically prescribed patterns of expected difficulty (e.g., detection tasks before production tasks, two-phoneme blends (e.g., $a-t$) progressing up sequentially to six-phoneme blending tasks (e.g., $s-p-r-i-$

n-t), and the early emergence of boundary consonants over medial vowels in word recognition and phonological tasks, the asymmetric later introduction of segmenting tasks, introduction of singleton letter sounds before complex digraphs, and so on). As the teachers became more familiar with ABRA during the session, they were encouraged to interact with the various online activities and to plan for student progression. Teachers were made aware that ABRA is only a tool, not an ICT “magic bullet.” They were shown that ABRA requires skilled teachers to implement it well and to link it effectively to cross-curricular learning outside the ABRA sessions. The concept of adaptive transformational use of technology as described earlier and reported in [Savage et al. \(2010\)](#) was discussed explicitly with all intervention teachers.

Once the intervention teachers had some hands-on exposure to the program, the investigators then presented and reviewed a suggested format for the teachers to use during a 1-hr ABRA lesson that specified 10 min of word-level work, 10 min of text-level work, 20 min of collaborative work, and 20 min of extension activities. The word-level work involved activities such as letter knowledge, phonological awareness, phonics, and word-building. Text-level work invites use of the fluency and comprehension activities based on the digital stories component in ABRA. For reading fluency, activities such as high-frequency words, reading with expression, reading accurately, and choral reading were suggested. For comprehension, activities that focused on predicting, comprehension monitoring, recognizing story elements, and summarizing as well as vocabulary and writing were identified. Emphasis was thus placed on demonstrating that ABRA acts as a balanced literacy program (e.g., [Pressley, 1998](#)).

All teachers were encouraged to use these activities but to select activities relevant to the appropriate point in children’s development. Thus, for kindergarten teachers, the phonic activities might be more likely to start with simpler activities within ABRA such as sound awareness, syllable and word counting, and aspects of rhyme awareness, whereas the teachers in Grade 1 might move their children more quickly to the simpler of the explicit blending tasks. Similarly for comprehension and aspects of the fluency tasks, teachers of kindergarten children were asked to encourage children to listen to stories and then complete tasks such as story ordering and summarizing, whereas teacher of children in Grade 1 were asked, where appropriate, to encourage children to read the texts and then complete comprehension tasks. This differentiated use of ABRA was also encouraged through the sustained in-class follow-up support for teachers.

Collaborative work encouraged students to work together in order to practice skills they would have learned in the earlier two sections. Collaborative work did not have to be conducted on a computer. This became an opportunity for students to work with and learn from each other. For example, we suggested that students could write alternative endings for the digital stories with a peer, engage in readers’ theatre, put on a puppet show, and so forth. *Extension activities* often included opportunities for the students to engage in collaborative work. For example, after reading *The Fruit Family* story, a kindergarten teacher could have her students draw pictures of the different types of fruit that they ate and label their pictures.

Teachers were informed that while regular access to appropriately leveled and progressively more demanding word-level and text-level activities were required, this suggested curriculum was a

fairly flexible guideline and should be adapted to meet the individual needs of their students as well as their own teaching styles. Teachers also had freedom to conduct the intervention as whole class, small group, individual, or some combination of the groupings. It was known at training that teaching would also vary depending upon access to technology in particular schools (e.g., presence of interactive whiteboard systems known as SMART Boards [SMART Technologies, Calgary, Alberta, Canada] or multiple computers in classrooms, use of a distinct ICT rooms in school). The responsibility for developing appropriate specific lesson plans and interventions always rested with the regular classroom teachers.

All of the teachers received a hard copy of an extensive ABRA teacher’s manual that illustrated in detail how ABRA could be used in these domains ([Abrami, White, & Wade, 2010](#)). Teachers also visited the Teacher’s Zone (available online at <http://abralite.concordia.ca/pd/index.php>), a resource area for them. Finally, the teachers met in small groups according to the grade levels they taught and planned ABRA introductory lessons for their classes based on the suggested 1-hr format.

Project management and roles. In each of the three provincial sites, an advanced graduate student coordinated the execution of the study under the supervision and support of a project literacy coordinator and a lead researcher in each province. Each provincial site’s co-coordinator was then responsible of recruiting and overseeing the training of their team of research assistants (RAs). The RAs visited participating classrooms and administered the pre- and posttest measures. During the first 4 weeks of the implementation phase, an RA was present in every experimental classroom to provide technical support and answer general pedagogical questions regarding the utilization of ABRA. Most teachers requested support for the first 3–5 lessons. The RAs also acted as the liaison between the teachers and the rest of the ABRA team. Monthly team meetings were held, and regular correspondence took place to update and address questions about data collection and lesson support. In the fifth week of intervention, the RAs started collecting treatment integrity data.

Literacy Assessment Measures

As ABRACADABRA was designed to aid children in learning alphabets, phonics, phonological awareness, reading fluency, oral language comprehension, and reading comprehension, reliable and valid psychometric tests were selected to examine all of these component abilities.

Letter-sound knowledge. To assess letter-sound knowledge, an experimenter showed a participant the 26 letters of the English alphabet and asked the student to say the corresponding sound of each letter presented following the assessment and scoring system described by [Savage et al. \(2009\)](#). The Spearman–Brown split-half internal reliability (*r*) of this test in the present sample at pretest was .87.

Blending words. This measure assesses a child’s phonological blending ability. A subtest of the Comprehensive Test of Phonological Processing (CTOPP) was used to examine students’ ability to blend words ([Wagner, Torgesen, & Rashotte, 1999](#)). In this test, the children listen to a series of disjointed sounds and then blend the sounds together to make a whole word. The Spearman–

Brown split-half reliability coefficient for this measure in the present sample at pretest was .86.

Fry words. To assess the students' word reading skills, we adapted a test using words from the Fry's Instant Word List (Fry, Kress, & Fountoukidis, 2000). Twenty words were randomly selected from Fry's first 200 words. The same 20 words were used at pre- and posttest. Each of the selected 20 words was placed on individual index cards and shown one at a time to participants. The students read each word presented to them and received a point for each word correctly read. The maximum score for this test was 20. The Spearman-Brown split-half reliability of this test in the present sample at pretest was .89.

Group Reading Assessment and Diagnostic Evaluation (GRADE). The GRADE is a standardized, nationally normed instrument designed to be administered to either the whole class or individually (Williams, 2001). The GRADE is reported to have strong internal consistency (r s ranging from .95 to .99) and retest reliability ($r = .80$; Williams, 2001). Reviews of the GRADE (Fugate, 2003; McBride, Ysseldecke, Milone, & Stickney, 2010; Waterman, 2003) have concluded that this tool is a reliable and valid measure of early reading ability.

Listening Comprehension subtest of the GRADE. We used the Listening Comprehension subtest to assess the students' understanding of spoken language. Children are read sentences and then asked to select a picture from four choices that best illustrates the meaning of each sentence. The Spearman-Brown split-half internal reliability coefficient for this measure in our sample at pretest was .68.

GRADE Sentence Comprehension and Passage Comprehension subtests. We used these Grade subtests to assess reading comprehension skills. The GRADE technical manual asserts that the reading comprehension test measures metacognitive skills of previewing, predicting, clarifying, and summarizing, making it a good measure of the comprehension skills taught in ABRA. The *Sentence Comprehension subtest* is a cloze procedure whereby students read a sentence that has a missing word and then select one of four words that best fits the context. In the *Passage Comprehension subtest*, students read short passages of around 32 words and then select the correct multiple-choice response to best answer each question. The Spearman-Brown split-half internal reliability coefficient in our sample at pretest was .76 for the sentence comprehension and .69 for passage comprehension.

Phonemic Segmentation Fluency subtest. The Phonemic Segmentation Fluency subtest of the Dynamic Indicators of Basic Literacy Skills (DIBELS; Kaminski & Good, 1996) assesses students' ability to fluently break three- or four-phoneme words into their individual phonemes in 1 min. For example, if the RA read *ship*, students had to say /sh/ /l/ /p/. The Spearman-Brown split-half reliability coefficient for this measure in the present sample at pretest was .97.

Oral Reading Fluency and Retell Fluency subtests. In the DIBELS, Oral Reading Fluency (ORF) subtest measures students' ability to read a passage out loud for 1 min. The number of correct words per minute from the passage yields the oral reading fluency rate. Students who correctly read 10 or more words in the DIBELS ORF task were then administered the Retell Fluency measure. Here, students tell the examiner all they can remember about the

passage in 1 min. The number of words retold yields the Reading Retell Fluency score.

Testing Procedure

All participants completed the Letter-Sound Knowledge, Fry Words, Blending Words, and GRADE Listening Comprehension measures. Other measures examining higher level literacy skills (reading fluency and reading comprehension) were developmentally appropriate only for students in Grades 1 and 1/2 at posttest. Fluency and comprehension measures were not administered to any of the kindergarten students at either pre- or posttest.

All children were seen twice at both pre- and posttest for testing. The first session involved individual testing of children. Here the Fry Words, Letter-Sound Knowledge and CTOPP Blending tasks were administered. Additionally, at posttest only for the Grade 1 and Grade 1/2 children, the three DIBELS subtests were also administered. In the second session, all GRADE measures were administered using a whole-class group-testing approach for children in two provincial sites and were individually administered in a third provincial site.

Treatment Integrity

In this study, a theoretically underpinned treatment integrity measure was developed that focused on the implementation of ABRA in the classrooms, following recommendations for best practice in assessing treatment integrity (Mowbray, Holter, Teague, & Bybee, 2003). Pairs of RAs first independently observed and recorded activities in around 20% of ABRA intervention classes. After each RA independently recorded his or her observations, they compared their ratings for agreement. Most pairs agreed on 98% of content of all observed sessions.

Observations were then formally coded using the Implementation Fidelity Measure (IFM), which was based on a rubric reported by Savage et al. (2010). Savage et al. adapted a classification system from Sandholtz, et al.'s (1997) stage model of computer technology integration by teachers that we described earlier, namely, entry, adoption, and adaptation. The criteria specified under each category of ABRA implementation also closely match the first three of Sandholtz et al.'s model criteria described earlier. Our first aim was to identify adequate treatment integrity. Teachers were informed during their training that adequate treatment integrity would consist of 20 hr of exposure to ABRA for each child in the intervention condition and that adequately implemented ABRA lessons would show evidence of careful planning, differentiation, and progression (e.g., in task difficulty) over time. Our aim, however, was to identify not just compliance with treatment but also possible excellence in implementation of ICT in classroom pedagogy. Whereas entry and adoption reflect traditional concerns about weak versus compliant intervention, adaptation refers to compliant intervention (as measured by reaching the same criteria for faithful *adoption* of ABRA) but *also* to clearly evident use of technology to transform wider learning in the classrooms. Two additions to the ABRA IFM rubric included the addition of a control level for teachers who did not use any aspect of ABRA technology, and a fifth category labeled as *differentiated adaptation* to highlight transformative use of ABRA in teaching across multiple skill levels including both word-level reading skills and

higher text-level comprehension and fluency extension activities. This latter category was distinguished from adaptation, which involved the transformative use of technology in *one* domain of literacy, most often in word-level reading skills. In understanding the differentiated adaptation measure, it should be noted that the use of ABRA at both word-level and higher text-level comprehension and fluency levels was encouraged but was never presented to teachers as a requirement of faithful adoption of ABRA. See Figure 2 for details of levels of implementation fidelity.

To establish rater reliability, we conducted a test trial of the rubric among four raters on four completed classroom observations. From a total of 12 independent IFM ratings, 11 of the judgments were identical across the four raters, thus establishing a 92% rater-reliability for the IFM rubric. Subsequently, every observation checklist completed during this study ($n = 293$) was rated by two of these raters. The raters had perfect agreement on 263 of the 293 IFM observation checklists they categorized. This is an 89% interrater agreement (Cohen's $\kappa = .83, p < .001$). The raters had 100% agreement in their ultimate categorization of teachers' use of technology on the 0–4 IFM scale.

Assessment of the general pedagogical quality of classrooms. To obtain further information about the quality of teaching, irrespective of whether teachers were implementing ABRA, observations of nontechnology based on regular language arts teaching was undertaken in both experimental and control classrooms. Senior RAs with previous early literacy experience conducted

observations using the Early Literacy and Language Classroom Observation (ELLCO; (Smith, Dickinson, & Sangeorge, 2002). The ELLCO is a standardized instrument assesses both the global quality of the classroom (e.g., classroom climate, approach to management, classroom organization), as well as language, literacy, and curriculum. The ELLCO takes about 90 min to complete. A brief teacher interview was also a part of this measure. In one provincial site ($n = 36$ classrooms), pairs of RAs observed the same lesson, rated the classes, and then met after the lesson to come to a rating consensus about the quality of classroom supports for literacy. Due to the availability conflicts of qualified RAs with an early literacy background, one rater, rather than two, collected data on this measure in two of the three provincial sites. Finally, we asked all teachers to complete a literacy intervention questionnaire (LIQ) devised by the current authors to ask about salient aspects of teachers' implementation of ABRA and teachers' attitudes toward the ABRA technology.

Results

Preliminary Data Analyses

Prior to analyses, data were first collated into separate child- and classroom-level spreadsheets. Preliminary data analyses suggested that there was no marked kurtosis or skewness in the classroom-level attainment data. All classroom-level variables were within

ABRA – IMPLEMENTATION RUBRIC

CONTROL (0)	ENTRY (1)	ADOPTION (2)	ADAPTATION (3)	DIFFERENTIATED ADAPTATION (4)
◆ No implementation ◆ No aspect of ABRA was used	◆ Little to No evidence of teacher planning of ABRA ELA lessons ◆ Little to No evidence of teacher instructional guidance ◆ Little to No evidence of teacher monitoring students' use of ABRA ◆ No evidence of teacher's awareness of zone of proximal development as students instructed to all work on same activity level ◆ Minimal student exposure to ABRA activities ◆ ABRA exposure mainly unstructured lessons where students choose own activities (play-time / free-time) ◆ Occasional disruption and off-task behaviour of students ◆ Unclear about teacher and student's navigational comfort level with ABRA - OR ◆ Teacher frustration/discomfort with technology evident	◆ Basic evidence of teacher planning of ABRA lessons ◆ Basic evidence of teacher instructional guidance of ABRA ◆ Basic evidence of teacher monitoring students' use of ABRA ◆ Some evidence of teacher's awareness of zone of proximal development as students are instructed to move up task levels if too easy / completed or move back if too hard ◆ ABRA exposure evidence of structured lessons ◆ Little off-task behaviour of students ◆ Teacher and students appear comfortable with navigating through ABRA activities ◆ Some evidence of differentiated use of ABRA activities, but mainly within one skill level (i.e. Phonics / Word Level activities).	◆ Clear evidence of teacher planning of ABRA lessons. Teacher links planning and target setting according to students ability level ◆ Clear evidence of teacher providing appropriate instructional guidance / feedback while students on ABRA ◆ Clear evidence of teacher monitoring students' use of ABRA ◆ Clear evidence of teacher's awareness of zone of proximal development as students are instructed to move up task levels if too easy / completed or move back if too hard ◆ ABRA exposure – evidence of structured lessons ◆ Students are clearly engaged in the lesson ◆ Teacher & students are comfortable with navigating through ABRA activities ◆ Extension of ABRA – Some evidence of entry-level activities that extend skills explored in one domain of ABRA, usually Word level (i.e., Rhyme matching game; spelling words or simple sentences; playing BINGO, etc). ◆ Evidence of differentiated use of ABRA activities, at more than one skill level (i.e. Word Level, Text Level or Writing activities). ◆ Evidence of collaborative work & use of collaborative learning opportunities	Criteria as per level four with the addition of the following: ◆ Extension of ABRA – Clear evidence of extension activities that incorporate higher-level skills (i.e. Comprehension) that extend beyond simple WORD level activities. Examples such as, writing alternate story endings; journal entry reflections on ABRA story; creating a drama skit/puppet show based on ABRA story, etc. ◆ Teacher clearly differentiated use of ABRA across ALL FOUR suggested levels of implementation (i.e. Word Level, Text Level, Collaborative Work & Extension Activities) ◆ Teacher uses all collaborative learning opportunities – peer supported dialogues, different roles, reciprocal tutoring, etc.

Original Observer / Class / Date: _____

RATER: _____ DATE: _____

SCORE:

Figure 2. ABRA Implementation Fidelity Measure (IFM) rubric. ABRA = acronym for ABRACADABRA (A Balanced Reading Approach for Canadians Designed to Achieve Best Results for All); ELA = English language arts.

acceptable limits of normality for skewness and kurtosis, so no data transformations were undertaken. There was no strong evidence of outliers. At this point, no data were excluded from analysis. In the case of classroom-level data, there was some missing data. In eight kindergarten classrooms, Listening Comprehension subtests were not administered by error. Four teachers did not wish children to be assessed on the Reading Comprehension and Fluency measures. In each case, this data loss was at the school level so that paired intervention and control classes were lost, thus producing no unbalancing of the overall RCT design. Imputation of an entire classroom's set of scores is not possible, so analyses for Listening Comprehension are based on 66 paired classrooms (74 – 8) and the Reading Comprehension and Fluency are based on 46 (50 – 4) paired classrooms.

Student-level missing data. We inspected the pattern of student-level missing data using the SPSS Missing Value Analysis (MVA) package to consider the randomness and impact of missing data and to then impute missing values where most appropriate. Missing data represented less than 5% of the total data across all variables, and so most common procedures for dealing with missing values would yield similar results (Tabachnick & Fidell, 2007). Analyses were performed using conventional chi-square tests to contrast the proportion of missing versus present data in intervention versus control group conditions at pre- and posttests. There was no evidence of selective experimental mortality across conditions, $\chi^2(1) < 1$, *ns*, in all cases. Further analysis using MVA revealed full-sample child-level data were also missing completely at random (MCAR) for all variables, according to Little's MCAR test ($p > .05$ in all cases). Regression-based imputation procedures were selected, with reading pretest variables serving as predictors. The means of the fifth iteration were selected for analyses.

Main Data Analyses

The classroom-level means and standard deviations of all classroom-level attainment variables at pre- and posttest are pre-

sented in Table 1. Inspection of Table 1 shows signs of posttest advantage for the ABRA group on Letter-Sound Knowledge, Phonological Blending, Fry Words, and Listening Comprehension. Few clear signs of advantage were evident in the Grade 1 Reading Comprehension and Fluency measures.

Does a classroom-level cluster RCT effectiveness trial of ABRA yield significant advantages for intervention over control classrooms in early literacy at posttest? In this cluster RCT, random allocation of students took place at the classroom level, producing a nested design in which the likely contextual influence that classrooms may have on the achievement of the individual participants can be evaluated (e.g., Hox, 2010; Raudenbush & Bryk, 2002). Our data were thus first analyzed with hierarchical linear modeling (HLM) with randomized classroom as the unit of analysis. The final HLM models were built in standard bottom-up fashion from preliminary analyses with steps in HLM followed sequentially in order to yield the final models. Model 1 was an unconditional one-way analysis of variance (ANOVA) model with random effects and confirmed that there was classroom-level variance at pretest and posttest on attainment measures beyond variance attributable to pupils and that HLM was appropriate.

Subsequent ANCOVA models tested whether candidate covariates were significant and should be retained in the final model. An ANCOVA model was appropriate in this design as pretest attainment, and chronological age was always a significant covariate of its corresponding posttest measure. The final three-level hierarchical model, built on these tested assumptions, sought to establish whether the significant classroom-level variance on posttest attainment measures (after control for school-level shared variance at Level 3, pretest classroom-level attainment variance at Level 2, and pre- and posttest pupil level attainment variance and pupil chronological age at Level 1) was explained by the ELA with ABRA versus ELA without ABRA factor. Equations 1, 2, and 3

Table 1
Comparison of Raw Score Means, Standard Deviations, and Adjusted Means by Experimental Condition

Measure	Intervention groups					Control group				
	Pretest		Posttest			Pretest		Posttest		
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	Adjusted ^a <i>M</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	Adjusted ^a <i>M</i>
Kindergarten & Grade 1/2										
Letter-Sound Knowledge	16.01	6.24	21.26	4.17	21.44	16.57	6.76	20.63	4.81	20.45
Fry Words	5.90	5.10	11.43	6.09	11.80	6.85	5.81	11.53	6.64	11.17
CTOPP Blending Words	6.41	2.94	9.96	3.40	10.27	7.19	3.04	9.48	3.56	9.17
DIBELS Phonemic Segmentation	—	—	28.75	11.33	29.78 ^b	—	—	27.60	11.41	26.57 ^b
GRADE Listening Comprehension	14.14	1.40	15.54	1.05	15.60	14.58	1.22	15.57	1.03	15.41
Grade 1/2										
DIBELS Reading Fluency	—	—	30.72	17.85	33.48 ^c	—	—	38.32	21.95	35.57 ^c
DIBELS Reading Retell	—	—	10.83	7.23	11.84 ^c	—	—	12.39	8.05	11.38 ^c
GRADE Sentence Comprehension	—	—	8.33	3.40	8.85 ^c	—	—	9.26	3.68	8.73 ^c
GRADE Passage Comprehension	—	—	9.00	3.44	9.57 ^c	—	—	9.94	3.95	9.38 ^c

Note. CTOPP = Comprehensive Test of Phonological Processing; DIBELS = Dynamic Indicators of Basic Literacy Skills; GRADE = Group Reading Assessment and Diagnostic Evaluation.

^a Adjusted means calculated using pretest and chronological age means as covariates. ^b Calculations for Phonemic Segmentation derived from pretest Blending Words mean. ^c Adjusted means for all Grades 1 & 2 reading measures (Reading Fluency, Reading Retell, Sentence Comprehension, & Passage Comprehension) calculated with the Fry Words pretest mean scores.

describe this final model at the pupil, classroom, and school levels, or student i in classroom j in school k , respectively.

Equation for Student Level-1 Model: (1)

$$Y_{ijk} = \pi_{00k} + \pi_{1jk}(\text{PRETEST}_{ijk}) + \pi_{2jk}(\text{CHRONOLOGICAL AGE}_{ijk}) + e_{ijk}$$

Equation for Classroom Level-2 Model: (2)

$$\pi_{00k} = \beta_{00k} + \beta_{01k} * (\text{PRETEST ATTAINMENT}_{jk}) + \beta_{02k} * (\text{INTERVENTION}_{jk}) + r_{0jk}$$

Equation for School Level-3 Model: (3)

$$\beta_{00k} = \gamma_{000} + \mu_{00k}$$

In these analyses, predictor variables were left uncentered and ratio-level raw scores which have a meaningful zero point value were used so as to ease interpretation. Equations 1–3 also show that the slope coefficients for all independent variables are treated as fixed: They are not allowed to randomly vary across classrooms and schools.

The results of these analyses are reported in Table 2. The measures administered to both kindergarten and Grades 1 and 2 are reported in Section A of Table 2. This analysis is of four measures—Letter-Sound Knowledge, Fry words, Phonological Blending, and Listening Comprehension—across all 74 paired classrooms (66 in the case of Listening Comprehension). These

Table 2
Hierarchical Linear Models Results for the Effect of ABRA Condition on Posttest Attainment

Variable	Student-level model		Classroom-level model	
	Coefficient	SE	Coefficient	SE
Section A ($n = 74$) Kindergarten & Grade 1/2 classrooms				
CTOPP Phonological Blending 2 = Dependent variable				
Chronological age	0.043	0.014**		
CTOPP Phonological Blending 1			0.631	0.024**
Intervention condition			1.080	0.320**
Letter-Sound Knowledge 2 = Dependent variable				
Chronological age	0.054	0.019**		
Letter-Sound Knowledge 1			0.518	0.018**
Intervention condition			0.961	0.461*
Fry Words 2 = Dependent variable				
Chronological age	0.099	0.021**		
Fry Words 1			0.629	0.022**
Intervention condition			0.691	0.690
DIBELS Phonemic Segmentation 2 = Dependent variable				
Chronological age	−0.308	0.062		
DIBELS Phonological Blending 1			1.508	0.109**
Intervention condition			2.578	1.425†
GRADE Listening Comprehension 2 = Dependent variable				
Chronological age	−0.007	0.008		
GRADE Listening Comprehension 1			0.365	0.023**
Intervention condition			0.085	0.160
Section B ($n = 50$) Grade 1/2 classrooms				
DIBELS Reading Fluency 2 = Dependent variable				
Chronological age	−0.177	0.153		
Fry Words 1			3.658	0.133**
Intervention condition			−2.517	2.282
DIBELS Reading Retell 2 = Dependent variable				
Chronological age	.119	.070*		
Fry Words 1			1.032	0.062**
Intervention condition			−0.148	0.969
Section B ($n = 46$) Grade 1/2 classrooms				
GRADE Sentence Comprehension 2 = Dependent variable				
Chronological age	−0.032	0.028	0.584	0.025**
Fry Words 1			0.148	0.375
Intervention condition				
GRADE Passage Comprehension 2 = Dependent variable				
Chronological age	0.012	0.031		
Fry Words 1			0.512	0.027**
Intervention condition			−0.027	0.440

Note. ABRA = shortened version of the acronym ABRACADABRA (A Balanced Reading Approach for Canadians Designed to Achieve Best Results for All) software program; CTOPP = Comprehensive Test of Phonological Processing; DIBELS = Dynamic Indicators of Basic Literacy Skills; GRADE = Group Reading Assessment and Diagnostic Evaluation.

† $p < .1$. * $p < .05$. ** $p < .01$.

results show that there is a significant effect of ABRA on intervention mean versus control group mean contrasts for Phonological Blending (10.27 vs. 9.17, $\beta = 1.08$, $p < .01$) and Letter-Sound Knowledge (21.44 vs. 20.45, $\beta = 0.96$, $p < .05$), and marginally for Phoneme Segmentation Fluency (29.78 vs. 26.57, $\beta = 2.58$, $p < .1$). Other effects however did not reach significance. Turning to the six measures of reading comprehension and fluency administered only to children in Grades 1 and 2 (50 paired classrooms), results showed no overall significant effects of ABRA on these measures.

Do high-implementing classrooms yield additional effects for ABRA? Prior to analyses, inspection of the IFM data revealed that of the 37 ABRA intervention teachers, there were seven entry-level teachers, 20 adoption-level teachers, seven adaptation-level teachers, and three differentiated-adaptation-level teachers. Due to the very small size of the last Time 1 (T1) category, the adaptation-level and differentiated-adaptation-level teachers were combined into a single adaptation-level category with 10 teachers. Of the entry-level teachers, six were kindergarten teachers, and one was a Grade 1 teacher. The means, standard deviations, and effect sizes by T1 condition are reported in Table 3 with the analyses involving the whole sample reported in Section A and those involving the Grade 1/2 in Section B of the table. There was evidence of value-added effect sizes for Phonological Blending, Letter-Sound Knowledge, and Phoneme Segmentation Fluency in the adoption and adaptation conditions relative to the control and entry conditions.

Our research questions here were specific and directional: We predicted largest improvements in literacy measures in adoption classrooms and high-implementing adaptation ABRACA-DABRA classrooms relative to control classrooms. Three-level (school, classroom, and pupil) HLM covariance models parallel to those used in the main analysis described earlier were used here. The only difference was that in these analyses, treatment integrity was considered as a factor entered to explore the relative differences in reading-dependent variable measures, with the control group as the reference category:

Equation for Student Level-1 Model: (4)

$$Y_{ijk} = \pi_{00k} + e_{ijk}$$

Equation for Classroom Level-2 Model: (5)

$$\pi_{00k} = \beta_{00k} + \beta_{01k} * (\text{ENTRY}_{jk}) + \beta_{02k} * (\text{ADOPTION}_{jk}) + \beta_{03k} * (\text{ADAPTATION}_{jk}) + r_{0jk}$$

Equation for School Level-3 Model: (6)

$$\beta_{00k} = \gamma_{000} + \mu_{00k}$$

Each Level-2 predictor was treated as fixed and not allowed to randomly vary across schools.

As Table 3 shows, the predicted scores for each of the three groups (entry vs. control, adaptive vs. control, and adoptive vs. control) were then used in subsequent planned comparisons. These analyses revealed that the adoption group had significantly higher mean performance than the control group in Phonological Blend-

ing (10.52 vs. 9.48, $z = 2.55$, $p = .01$), Phoneme Segmentation Fluency (31.81 vs. 26.94, $z = 3.07$, $p < .01$), Sight Word Reading (12.87 vs. 11.26, $z = 1.99$, $p < .05$), and marginally for Letter-Sound Knowledge (21.46 vs. 20.69, $z = 1.41$, $p = .08$). The adaptation group similarly showed significantly higher performance than the control group on Letter-Sound Knowledge (22.58 vs. 20.69, $z = 2.37$, $p < .01$), Phonological Blending (10.70 vs. 9.48, $z = 1.97$, $p < 0.01$), and Phoneme Segmentation Fluency (31.30 vs. 26.94, $z = 1.85$, $p < .05$). No advantages were evident in any analysis for the entry group. This analysis confirms the presence of comparable significant advantages over controls for both the adoption and adaptation teachers who together made up 80% of the intervention teacher sample.

A final issue in the experimental study was the comparison of 74 intervention and control classrooms on background ELLCO measure four subscales—Language Literacy Curriculum, Classroom Observation, Book Reading, and Writing—and combined total score—Literacy Activities Overall Rating Scale total. The question this analysis addressed was whether ABRA intervention and control classrooms are comparable on the observed quality of their non-ABRA regular ELA teaching. The mean scores and results of these analyses are shown in Table 4. ANOVA analyses show that intervention and control group teachers are highly comparable both on overall ELLCO and on subscale measures, all F s < 1 , ns , except for the Writing subtest, where $F(1, 72) = 2.67$, $p = .11$. Analysis was undertaken on ELLCO scores across T1 category to explore whether the stronger implementers of ABRA were simply better teachers overall than the less-strong implementers. The results are reported in Table 5. These ANOVA analyses show that there were no significant differences in ELLCO score by implementation type, all F s < 1.45 , ns , except for the Writing subtest, where $F(2, 64) = 2.86$, $p = .07$, consistent with the view that the technology-implementers were not simply better literacy teachers overall.

Explorations of poor implementation fidelity. Beyond the experimental contrasts previously described, an important issue in any educational field study concerns understanding why teachers chose either not to implement the prescribed intervention or to implement it suboptimally. The issue is important because relevant evidence can guide future attempts to improve implementation and provides a sense of whether take-up of interventions is achievable. The present article carefully explored treatment integrity, focused on extended training of teachers, observed teachers' practice, and asked teachers about their practice through questionnaires and discussions. Detailed records of all of these elements were recorded to assess whether scale-up of the study was achievable and under what conditions.

What is known from this data of patterns of treatment integrity? First our direct observations reported in Table 3 showed that of 37 teachers asked to implement ABRA in this study, only seven were judged to have performed at the entry level of poor implementation. All 30 other teachers (80% of the intervention sample of teachers) achieved at least an adoption level of implementation or adopted the program faithfully as well as adapting the teaching in their classrooms, and as described earlier, both of these groups were associated with significant gains in a range of early literacy measures. As noted earlier, of the seven entry-level teachers, only one was a Grade 1 teacher, the other six all being kindergarten teachers. Thus, there was 96% adoption of ABRA among the

Table 3

Predicted Means, Standard Errors, and Adjusted Means by Treatment Quality Groups: Kindergarten and Grade 1/2 Measures

Measure/Treatment quality level	Pretest			Posttest			Adjusted <i>M</i>
	<i>n</i>	Mean	<i>SE</i>	<i>n</i>	Mean	<i>SE</i>	
Letter–Sound Knowledge							
Control	37	16.50	1.07	37	20.64	0.74	20.69 ^a
Entry	7	10.25	2.29	7	17.31	1.07	21.03 ^a
Adoption	20	18.64	1.49	20	22.82	2.43	21.46 ^a
Adaptation	10	14.57	2.17	10	21.33	4.30	22.58 ^a
Fry Words							
Control	37	6.91	0.99	37	11.53	6.64	11.26 ^a
Entry	7	2.49	1.89	7	6.29	2.26	9.85 ^a
Adoption	20	7.96	1.22	20	13.88	1.47	12.87 ^a
Adaptation	10	5.03	1.81	10	11.16	2.16	11.94 ^a
CTOPP Blending Words							
Control	37	7.18	0.55	37	9.70	0.67	9.48 ^a
Entry	7	4.14	1.03	7	7.77	1.16	9.82 ^a
Adoption	20	7.71	0.66	20	11.17	0.74	10.52 ^a
Adaptation	10	6.17	0.99	10	10.19	1.11	10.70 ^a
DIBELS Phonemic Segmentation							
Control	—	—	—	37	27.61	1.93	26.94 ^b
Entry	—	—	—	7	16.84	3.88	23.41 ^b
Adoption	—	—	—	20	33.63	2.51	31.81 ^b
Adaptation	—	—	—	10	30.84	3.72	31.30 ^b
GRADE Listening Comprehension							
Control	33	14.54	0.25	33	15.58	0.20	15.51 ^c
Entry	6	14.15	0.49	6	15.74	0.41	15.83 ^c
Adoption	19	13.91	0.28	19	15.41	0.24	15.59 ^c
Adaptation	9	13.71	0.44	9	15.12	0.37	15.33 ^c
DIBELS Reading Fluency							
Control	—	—	—	25	38.95	4.33	29.55 ^c
Entry	—	—	—	6	8.15	14.24	34.96 ^c
Adoption	—	—	—	16	31.84	4.51	23.75 ^c
Adaptation	—	—	—	8	25.84	6.47	26.87 ^c
DIBELS Reading Retell							
Control	—	—	—	25	12.91	1.71	8.91 ^c
Entry	—	—	—	—	5.02	5.70	11.25 ^c
Adoption	—	—	—	16	11.11	1.80	7.88 ^c
Adaptation	—	—	—	8	10.65	2.58	9.82 ^c
GRADE Sentence Comprehension							
Control	—	—	—	23	9.28	0.81	7.76 ^c
Entry	—	—	—	—	5.85	2.09	9.48 ^c
Adoption	—	—	—	14	8.90	0.72	7.89 ^c
Adaptation	—	—	—	8	6.61	0.81	6.77 ^c
GRADE Passage Comprehension							
Control	—	—	—	23	10.28	0.82	8.61 ^c
Entry	—	—	—	6	4.51	2.17	7.61 ^c
Adoption	—	—	—	14	10.44	0.74	9.36 ^c
Adaptation	—	—	—	8	7.00	1.00	6.85 ^c

Note. The covariate measures used in calculating the adjusted mean values for the reading measures were the pretest Fry Words and chronological age of students at pretest. CTOPP = Comprehensive Test of Phonological Processing; DIBELS = Dynamic Indicators of Basic Literacy Skills; GRADE = Group Reading Assessment and Diagnostic Evaluation.

^a Adjusted means calculated using corresponding pretest and chronological age as covariates. ^b Adjusted means for DIBELS Phonemic Segmentation calculated using pretest CTOPP Blending Words and chronological age value of students as covariates. ^c Adjusted means calculations for all Grades 1 and 2 reading measures (Fluency, Retell, Sentence Comprehension, & Passage Comprehension) used the Fry Words pretest scores.

Grade 1 or Grade 1/2 teachers, but only 50% adoption among kindergarten teachers in our sample, a by-grade difference that was highly significant, $\chi^2(1) = 14.30, p = .0002$.

On the other hand, it was very rare to find teachers in any grade in this study who adapted their teaching with ABRA to transform wider classroom teaching in both of the domains of word-level analysis and text-level comprehension. Indeed the very small numbers here (three from 37 or less than 10% of implementing teachers) meant that a combined measure of

adapted teaching was used to describe any ABRA teaching that transformed any aspect of classroom literacy instruction. Of the three teachers who did use such a transformative approach, it may have been that they had greater expertise in both literacy and technology to allow this to happen. In at least one case of the three, for example, the teacher had recently returned to the classroom after operating as a school board literacy consultant and was highly familiar with white board technologies in education.

Table 4

One-Way Analysis of Variance for Differences Between ABRA Condition Groups as a Function of Early Literacy and Language Classroom Observation (ELLCO) Measure Subtotals

ELLCO measure subtotals	Intervention group (<i>n</i> = 37)		Control group (<i>n</i> = 37)		ANOVA differences between ABRA condition groups		
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>df</i>	<i>F</i>	<i>p</i>
Language literacy curriculum	31.81	4.67	31.51	5.12	1	0.07	.80
Classroom observation	55.22	7.99	54.68	8.73	1	0.08	.78
Book reading	5.22	1.80	5.16	1.63	1	0.02	.89
Writing	3.76	.98	3.38	1.01	1	2.67	.11
Literacy activities overall rating scale total	8.97	1.92	8.54	1.88	1	0.96	.33

Note. ANOVA = analysis of variance; ABRA = shortened version of the acronym ABRACADABRA (A Balanced Reading Approach for Canadians Designed to Achieve Best Results for All) software program.

Analyses of the ELLCO scores (performed with one-sample *t* tests) of the one entry-level Grade 1 teacher against the 25 other intervention Grade 1 teachers showed that this teacher achieved significantly lower ELLCO scores on four of the five ELLCO measures than their peers, $t(25) = 3.36$, $p = .003$ for Language, Literacy, and Curriculum; $t(25) = 4.57$, $p < .001$ for Classroom Observation total; $t(25) = 4.05$, $p < .001$ for Book Reading;

$t(25) = -1.37$, *ns*, for Writing, and $t(25) = 2.97$, $p = .007$ for Literacy Activities Rating Scale total. By contrast, the six entry-level kindergarten teachers were not so rated, compared with their kindergarten teachers peers in univariate ANOVAs contrasting them with the six adoption or adaptation teachers, $F(1, 10) < 1$, *ns*, for Language, Literacy, and Curriculum; $F(1, 10) = 1.46$, *ns*, for Classroom Observation total; $F(1, 10) = 2.11$, *ns*, for Book Read-

Table 5

One-Way Analysis of Variance for Differences Among ABRA Implementation Fidelity Measure Condition Groups as a Function of ELLCO Measure Subtotals

ELLCO measure subtotals/treatment integrity level	<i>n</i>	<i>M</i>	<i>SD</i>	ANOVA differences between treatment integrity groups		
				<i>df</i>	<i>F</i>	<i>p</i>
Language literacy curriculum				4	1.28	.29
Control	37	31.51	5.12			
Entry	7	29.14	5.90			
Adoption	20	32.80	4.36			
Adaptation	7	33.14	2.97			
Differentiated	3	31.66	4.87			
Classroom observation				4	1.43	.23
Control	37	54.68	8.73			
Entry	7	49.43	8.60			
Adoption	20	57.20	7.91			
Adaptation	7	57.00	4.55			
Differentiated	3	51.33	9.07			
Book reading				4	1.13	.35
Control	37	5.16	1.63			
Entry	7	4.00	1.63			
Adoption	20	5.50	1.57			
Adaptation	7	5.57	2.44			
Differentiated	3	5.33	1.53			
Writing				4	2.17	.08
Control	37	3.38	1.01			
Entry	7	4.14	.69			
Adoption	20	3.40	1.10			
Adaptation	7	4.14	.69			
Differentiated	3	4.33	.58			
Literacy activities overall rating scale total				4	.94	.44
Control	37	8.54	1.88			
Entry	7	8.14	1.35			
Adoption	20	8.90	1.97			
Adaptation	7	9.71	2.36			
Differentiated	3	9.67	1.53			

Note. ABRA = shortened version of the acronym ABRACADABRA (A Balanced Reading Approach for Canadians Designed to Achieve Best Results for All) software program; ELLCO = Early Literacy and Language Classroom Observation; ANOVA = analysis of variance.

ing; $F(1, 10) = 1.60$, *ns*, for Writing; and $F(1, 10) < 1$, *ns*, for Literacy Activities Rating Scale total. The six kindergarten entry-level teachers were also equally distributed across the three provinces of this study, and none were beginner teachers.

We inspected data from our LIQ teacher questionnaire. Data were available for three of the six entry-level kindergarten teachers, the one Grade 1 entry-level teacher, and four of six adoption- or adaptation-level kindergarten teachers. Responses to questions concerning the effects of technological difficulties did not show that entry-level teachers were more likely to experience technical difficulties implementing ABRA. Overall, half of both the entry- and adoption-level teachers commented on this issue as a factor. Teachers did not report that their knowledge of software was negatively reflected in reduced use of ABRA. Indeed, none of the entry-level teachers suggested this was an issue, whereas two of the four adoption- or adaptation-level teachers mentioned it as being an issue on occasion. These data are also consistent with a detailed analysis of the training of a subset of all teachers using ABRA in one site in this study reported by Anderson et al. (2011). Results of this study suggested dramatic reductions in requests for just-in-time assistance over the first 2 weeks or so of implementation of ABRA such that all teachers showed that they were well trained to use the technology competently and independently.

Finally, inspection of field notes from teacher training and discussions with staff leading the support for teachers were consistent with this general picture. Entry-level teachers were seen as being at least capable, and often excellent, teachers, but sometimes rather wary of technology or the use of any technology to teach reading in kindergarten on general pedagogical-philosophical grounds. At least some of these teachers drew attention to their provincial kindergarten curricula that did not encourage explicit phonological analysis of words or the teaching of reading at this early stage of children's careers. In sum, the majority of teachers (80% of intervention teachers) used ABRA with fidelity, and nearly all (96%) of Grade 1 teachers did so. Of the 20% who did not do so, the clearest feature of the group was that they were more likely to be kindergarten teachers, and while strong teachers, they sometimes articulated that on general pedagogical or curricular grounds that technology was not suitable or that the skills in ABRA were not suitable for children in kindergarten. On the other hand, even in kindergarten an equally large number of teachers used ABRA with fidelity, and as our analyses show, all such teaching showed significant advantages on a range of literacy measures.

Discussion

The main aim of the present study was to undertake a well-designed cluster RCT trial to evaluate the added value of ABRA delivered by regular classroom teachers in their regular kindergarten through Grade 2 ELA classes as a test of external validity of the ABRA resource. We sought to answer two questions: the first question concerned the overall effects of ABRA in intervention versus control group contrasts, and the second question concerned the relative effects of different levels of treatment integrity on outcomes.

Did ABRA Yield Significant Advantages for Intervention Over Control Classrooms in Early Literacy at Posttest?

Our analyses found that ABRA produced significant effects on standardized measures of Phonological Blending at posttest in 37 classrooms representing different school boards, regions, and educational policies from across Canada compared with 37 comparison ELA classrooms nested within the same schools that did not use ABRA between pre- and posttest. There were also significant effects of ABRA on Letter-Sound Knowledge and marginally significant effects on speeded phoneme segmentation ability. This specific pattern of effects of ABRA on phonological abilities replicates patterns reported in four previous studies of ABRA (Comaskey et al., 2009; Savage et al., 2009; Wolgemuth et al., 2011, 2012). The results of the present study are an important novel contribution to knowledge as this is the first time that an *effectiveness* study, that is, a study conducted by regular classroom teachers rather than an *efficacy* study conducted by research students or specially employed and trained educators, has shown ABRA to improve these aspects of literacy. Against this finding, and contrary to hypotheses, there were no significant effects of ABRA intervention on measures of Sight Word Reading, listening and reading comprehension, and fluency.

Theoretically, the development of phonological blending skills is closely allied to the ability to "decode" novel words. This ability has been described as the sine qua non of reading acquisition in alphabetic orthographies (e.g., Share, 1995, 2008). A large amount of evidence from experimental, longitudinal, and intervention research shows that children with strong blending and decoding skills have a greater likelihood of going on to be fluent readers and comprehenders of text than those with weak skills in these domains (e.g., Byrne, 1998; National Reading Panel, 2000; Savage, Carless, & Ferrero, 2007; Wagner et al., 1999). For these reasons, letter-sound knowledge, phonological blending skills, and decoding have appropriately been called foundations of literacy (e.g., Byrne, 1998; Seymour, 1997).

There were no significant effects for two other measures administered across the sample: Sight Word Reading and Listening Comprehension. There were also no significant effects for ABRA across five measures of comprehension and fluency, most of which were administered to children in Grade 1. It might be that the significant impact of ABRA on phonological awareness and letter knowledge at immediate posttest is reflected in significant growth in reading comprehension at a delayed posttest. Indeed, recent data suggest this pattern (Di Stasio et al., 2012). Nevertheless, the pattern here is somewhat disappointing, given that ABRA is designed to be a balanced literacy resource. There are a number of possible explanations for the patterns in the present study. One concern is that the smaller sample for these latter tests (48 classrooms) and the lower internal reliability of the comprehension tests may have impacted our ability to assess outcomes. However, the absence of trends favoring the intervention condition in sample estimates argues against the former interpretation. Alternatively, it is possible that ABRA does not implement comprehension and fluency strategies as well as it does phonic strategies. However, the finding that ABRA significantly improves comprehension and fluency in efficacy trials conducted by university researchers but not in teacher-led effectiveness trials argues against this view. A

more direct explanation of different patterns lies in the implementation of comprehension in teacher-led ABRA interventions. This issue of implementation is considered below.

Do Adequate and Transformational Implementations Yield Additional Effects for ABRA?

The second research question explored in the present study concerned the impacts of different levels of quality in implementation of ABRA. Detailed investigation of treatment integrity are rarely undertaken or reported fully (e.g., Campuzano et al., 2009; Lane et al., 2004). Detailed records were kept of 293 paired classroom observations of teachers using ABRA in the present study, yielding high interrater reliability for the coding system constructed. Further analysis of student data treating variation in the observed quality of teaching showed that ABRA effects were, as predicted, partly mediated by the quality of technology implementation by teachers. Teachers who were entry-level implementers with inadequate exposure to ABRA, poorer evidence of planning, differentiation, and progression in learning produced little growth in student literacy compared with the majority of teachers whose implementations were of superior quality at the adoption or adaptation levels of implementation. There was little evidence in size or extent of effects of an additional advantage for adaptation over adoption. This finding suggests that faithful adherence to delivery of the content of ABRA is the main source of the significant growth in literacy described here.

As noted earlier, there is much evidence that the quality of the implementation of programs has a significant impact on outcomes regardless of whether the interventions are successful and involve technology (Chambers, Abrami, et al., 2008; Savage et al., 2010; Wolgemuth et al., 2011; Wozney et al., 2006) or do not involve technology (e.g., Davidson et al., 2009; Lane et al., 2004; Stein et al., 2008). The current findings are consistent with this literature. However, it should be noted that unlike some studies (e.g., Chambers, Abrami, et al., 2008; Davidson et al., 2009), variation in program implementation did not *entirely* explain implementation effects. That is, overall significant effects of ABRA on phonic skills were evident across the whole sample irrespective of implementation quality. This is important as it shows that ABRA is effective across samples of teachers, given current abilities, while the treatment integrity effects suggest that if the largely kindergarten-based entry-level teachers can be supported to integrate or embed technology as skillfully as the adoption-level and adaptation-level teachers did, the impact of technology on literacy may be still greater still. Our analysis of these teachers' implementation suggests the need to win a wider debate about the suitability of educational technology in kindergarten and to be aware that local curricular constraints can influence uptake of technologies. In this sense, teacher professional development as well as the development of provincial curricula that fully support technology use may be required for maximal treatment integrity.

The same treatment integrity measure also showed that there were only three (two Grade 1/2 and one kindergarten classroom) teachers in the whole sample of 37 intervention teachers who adapted ABRA for both word-level decoding and word reading and text-level comprehension and fluency work. This finding was reported despite the fact that a substantial amount of time was spent covering these four important elements of literacy acquisi-

tion during the teacher training and subsequent feedback sessions. While our definition of treatment integrity called for teachers to use ABRA as they saw best for 20 hr per child in their classrooms, deliberate efforts were made during training sessions to have teachers focus on activities that addressed higher order skills. For example, training often began with a focus on comprehension and fluency sections and then later dealt with alphabetics, so teachers would become familiar doing the "more difficult" comprehension activities first and then move on to the "easier" phonic skills. These views were then reiterated during the ongoing support and advice given to teachers. The finding that few teachers sought to use the comprehension components of ABRA provides both an obvious explanation of the null effects of ABRA on children's comprehension skill reported earlier and also suggests that further or more effective training must be done with teachers to incorporate the software within their classroom instruction.

In contextualizing these findings, it could be noted that there is relatively limited evidence that teachers use explicit strategy teaching for comprehension outside ICT (e.g., Pressley, 1998). It might be that in contrast to teaching phonics that, while not ubiquitous, is common in some form, teachers do not know how to explicitly teach cognitive strategies for comprehension. A second issue that should be borne in mind is that the ABRA intervention was conducted toward the beginning of the academic year with many beginner readers. It may be that teachers focus at this time on word reading and phonics, and balance is achieved in instruction not by concurrent equivalent emphasis (as required in this evaluation of ABRA) but rather by working more on comprehension later in the year. Certainly, recent observational studies in Grade 1 classes in some overlapping vicinities suggest that in the latter part of year, more of the focus is on comprehension and not exclusively on phonics (Deault, 2011; Deault & Savage, 2012), at least in Grade 1 classrooms.

Finally, it may be the case that the comprehension activities in ABRA do not go far enough in scaffolding the pedagogy of teachers and the learning of their students. Pressley (2002) concluded that there was little evidence of teachers encouraging students to become self-regulated readers by using good comprehension strategies. There was the expectation that students would be self-regulated in their reading, but no instruction of these self-regulated processes was given. Abrami, Lysenko, Wade, and Pillay (2011) linked the structured activities in ABRA with the self-regulation activities in ePEARL (LEARN, Laval, Quebec, Canada), an electronic portfolio tool. In a longitudinal quasi-experiment, Abrami et al. (2011) found positive effects on comprehension in medium- and high-implementation ABRA-ePEARL classrooms compared with control classrooms.

Turning to the wider literature on technology, as noted earlier, several comprehensive high-quality systematic reviews of RCT research suggest few positive effects of ICT on literacy (e.g., Andrews et al., 2007; Torgersen & Zhu, 2003). Much of the research included in these full historical reviews was undertaken in the 1980s and 1990s. More recently, high-quality RCT studies have been conducted around the world—for example, in France, (Ecalte et al., 2009), the United States (Chambers, Abrami, et al., 2008; Chambers, Cheung, Madden, Slavin, & Gifford, 2006; Chambers, Slavin, et al., 2008), Australia (with ABRA, Wolgemuth et al., 2011, 2012), and in Canada (with ABRA, Comaskey et al., 2009; Di Stasio et al., 2012; Savage et al., 2009). The current

ABRA research joins this small but growing literature. We thus concur with Andrews et al. (2007) who noted that it is a particularly exciting time for ICT research in literacy as both the theory of ICT and the evidence base drawn from this more sophisticated theoretical positioning is in its infancy. A more sophisticated theoretical position would also need to ask not just that technology works, but why it works.

Underlying Cause of the Effectiveness of ABRA

In a field-based effectiveness trial, the sources of influences on outcomes are, by definition, greater than in internal validity trials such as that reported by Savage et al. (2009). Effectiveness trials represent a distinct and crucial step in any evaluation of the utility of interventions. Why then did ABRA work here for phonic-related skills? Specifically, we have argued earlier that three overarching methodological issues are of the essence in high-quality ICT intervention: the *quality of the technology* used in research studies, the *quality of the implementation* of intervention studies, and the *theoretical and pedagogical coherence* of technologies. The quality of the ABRA technology, we argue, is high; for example, unlike most ICT, ABRA has a fully implemented synthetic phonics component (e.g., Grant et al., 2012). This may have supported the strong growth witnessed in phonic-related skills.

A second theme is that one needs to explore actions in the classroom in detail to understand the effects of implementation. Startlingly, some major intervention studies (e.g., Campuzano et al., 2009) do not explore implementation at all, so results tell us very little about an intervention. Another fact about implementation that might explain the effectiveness of ABRA is that ongoing and embedded support was provided to teachers. Unlike many training studies, here all teachers in this study participated in a training day and also received continuous support when delivering ABRA in their language arts classes. We gave constant support through regular visits and phone and e-mail advice in addition to detailed guidance on pacing the curriculum and feedback on teaching for the first 3–4 weeks. After that initial guidance, nearly all teachers no longer needed direct facilitator support to use ABRA. Our perception was that this initial direct support was crucial to the effective implementation of ABRA. Finally, we note that the Slavin, Lake, Chambers, Cheung, and Davis' (2009) systematic review of interventions suggested that the largest improvements found across intervention studies involve clear changes in teacher practice rather than changes to curriculum or other features of classroom life such as the use of ICT per se. For this reason alone, explorations of changes in teacher practice during and after interventions are of high importance.

The third issue concerns the theoretical and pedagogical coherence of technologies for literacy. We have been clear to articulate an embedded role for technology alongside of and highly connected to effective teaching in regular classrooms. Here technology can add value as an additional resource compared with effective teaching in the absence of technology. This model is of course reflected in our statistical analyses, where the value added by ABRA to classrooms was the focus. For this reason, the clinical trial model, while important, cannot be too rigidly applied as the best teaching with ICT is likely to be highly interconnected with other forms of learning. We have also pointed to a range of evidence sources that suggests that an implementation science of

reading needs to focus on what teachers do and also why they do and do not do things, and on the cultural aspects of the profession as much on interventions themselves. In ABRA specifically, there might thus be a role for exploring through “thick description,” the technology-based and nontechnology-based aspects of their classroom teaching before and well after ABRA has been introduced into classrooms.

Limitations of the Present Study

A number of potentially important potential limitations need to be noted in the present study. One issue concerns the relatively wide range in chronological age. There were, by chance, more children in Grade 2 in the control condition than in the intervention condition. Stratified randomization was not possible in small schools with one Grade 1 class and one Grade 1/2 class or larger ones with uneven numbers of classes at each grade level. It was judged to be methodologically superior to randomize within schools to control for the well-documented school-level effects on attainment, rather than to stratify across schools for age. The effects of these differences in age were probably modest: there was no significant difference between the intervention and control groups across the sample in chronological age, and effects reported persisted after control for chronological age. The results are potentially limited by reporting effects at immediate posttest and not also at delayed posttest. However, a precondition for obtaining the capacity to randomize interventions in schools was our promise to train the control class teachers after the intervention period was completed. As noted earlier, much longitudinal research, including our own work with ABRA, strongly suggests that the phonological and letter skills that improved here are foundational for later literacy. The patterns reported at posttest (e.g., significant effects for phonological and letter knowledge tasks overall and enhanced effects for high implementers) were also very closely related to observed teacher behaviors with ICT showing that gains observed are unlikely to be solely of a general motivational nature.

Implications and Future Studies

Arguably, the single biggest challenge facing reading researchers, implementation scientists, and practitioners in the 21st century is the issue of building scalable and sustainable interventions (Abrami et al., 2008). We demonstrated here that ABRA, an open, free-access resource, can be used effectively at scale across Canada. It has also used efficaciously in remote regions of northern Australia (Wolgemuth et al., 2011, 2012). ABRA has also been embedded within the practice of several school boards, ABRA's professional support staff, and others around the world. ABRA is thus a community resource in the long-established spirit of giving away the best and most useful findings of scientific psychology (Miller, 1969).

The present research also shows that such studies are possible for university-based teams to execute successfully in Canada. The culture of funding, collaborating between and in schools, and executing large-scale RCT studies is not at all widespread in Canada (e.g., Jamieson, 2006). Indeed, this study is the largest non-U.S. study of ICT to date. Indeed, as far as we are aware, this is largest true experiment (RCT) reading intervention of any sort reported to date in Canada. We would argue that this sort of work

yields clear value-added dividends for the community. The next step must be to explore the scale-up of interventions, to obtain a richer picture of the support that teachers need to encourage both high-level adaptations and strong expectancies of success, and to determine the long-term effects of intervention, ways to use evidence to design better and more effective ABRA activities, and how to best hand-over ABRA effectively to school boards. All of this is work in progress.

References

- Abrami, P. C., Lysenko, L., Wade, A., & Pillay, V. (2011). *A quasi-experimental study of the use of literacy and portfolios tools to support the comprehension skills of emerging readers*. Montreal, Quebec, Canada: Concordia University Centre for the Study of Learning and Performance.
- Abrami, P. C., Savage, R. S., Deleveaux, G., Wade, A., Meyer, E., & Lebel, C. (2010). The Learning Toolkit: The design, development, testing and dissemination of evidence-based educational software. In P. Zemliansky & D. M. Wilcox (Eds.), *Design and implementation of educational games: Theoretical and practical perspectives* (pp. 168–188). Hershey, PA: IGI Global. doi:10.4018/978-1-61520-781-7.ch012
- Abrami, P. C., Savage, R. S., Wade, C. A., Hipps, G., & Lopez, M. (2008). Using technology to assist children learning to read and write. In T. Willoughby & E. Wood (Eds.), *Children's learning in a digital world* (pp. 129–171). Oxford, England: Blackwell. doi:10.1002/9780470696682.ch6
- Abrami, P. C., White, B., & Wade, A. (2010). *ABRACADABRA LTK teacher guide*. Retrieved from http://grover.concordia.ca/abracadabra/resources/download/LTK_ABRA_lr.pdf
- Anderson, A., Wood, E., Piquette-Tomei, N., Savage, R. S., & Mueller, J. (2011). Evaluating the impacts of just-in-time instructional support for teachers introducing a web-based reading program for primary grade children. *Teaching and Teacher Education*, 19, 499–525.
- Andrews, R., Freeman, A., Hou, D., McGuinn, N., Robinson, A., & Zhu, J. (2007). The effectiveness of information and communication technology on the learning of written English for 5- to 16-year-olds. *British Journal of Educational Technology*, 38, 325–336. doi:10.1111/j.1467-8535.2006.00628.x
- Bereiter, C. (2002). *Education and mind in the knowledge age*. Mahwah, NJ: Erlbaum.
- Bishop, D. V. M., & Snowling, M. J. (2004). Developmental dyslexia and specific language impairment: Same or different? *Psychological Bulletin*, 130, 858–886. doi:10.1037/0033-2909.130.6.858
- Blok, H., Oostdam, R., Otter, M. E., & Overmaat, M. (2002). Computer-assisted instruction in support of beginning reading instruction: A review. *Review of Educational Research*, 72, 101–130. doi:10.3102/00346543072001101
- Borenstein, M., Rothstein, H., Cohen, J., Schoenfeld, D., & Berlin, J. (2001). Power and precision (Version 2.1) [Computer software]. Englewood, NJ: Biostat.
- Byrne, B. (1998). *The foundation of literacy: The child's acquisition of the alphabetic principle*. Hove, England: Psychology Press.
- Campuzano, L., Dynarski, M., Agodini, R., Rall, K., & Pendleton, A. (2009). Effectiveness of reading and mathematics software products: Findings from two student cohorts (NCEE 2009–4041). Jessup, MD: National Center for Education Evaluation and Regional Assistance. Retrieved from <http://ies.ed.gov/ncee/pubs/>
- Chambers, B., Abrami, P., Tucker, B., Slavin, R. E., Madden, N. A., Cheung, A., & Gifford, R. (2008a). Computer-assisted tutoring in Success for All: Reading outcomes for first graders. *Journal of Research on Educational Effectiveness*, 1, 120–137. doi:10.1080/19345740801941357
- Chambers, B., Cheung, A. C. K., Madden, N. A., Slavin, R. E., & Gifford, R. (2006). Achievement effects of embedded multimedia in a success for all reading program. *Journal of Educational Psychology*, 98, 232–237. doi:10.1037/0022-0663.98.1.232
- Chambers, B., Slavin, R. E., Madden, N. A., Abrami, P. C., Tucker, B. J., Cheung, A., & Gifford, R. (2008b). Technology infusion in Success for All: Reading outcomes for first graders. *The Elementary School Journal*, 109, 1–15. Retrieved from http://successforall.org/_images/pdfs/Technology_Infusion_11_04_05.doc. doi:10.1086/592364
- Chen, C. H. (2008). Why do teachers not practice what they believe regarding technology integration? *The Journal of Educational Research*, 102, 65–75. doi:10.3200/JOER.102.1.65-75
- Chen, J. Q., & Chang, C. (2006). Using computers in early childhood classrooms. *Journal of Early Childhood Research*, 4, 169–188. doi:10.1177/1476718X06063535
- Clark, J. M., & Paivio, A. (1991). Dual coding theory and education. *Educational Psychology Review*, 3, 149–210. doi:10.1007/BF01320076
- Clarke, D., & Hollingsworth, H. (2002). Elaborating a model of teacher professional growth. *Teaching and Teacher Education*, 18, 947–967. doi:10.1016/S0742-051X(02)00053-7
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Hillsdale, NJ: Erlbaum.
- Coltheart, M., Curtis, B., Atkins, P., & Haller, M. (1993). Models of reading aloud: Dual-route and parallel-distributed-processing approaches. *Psychological Review*, 100, 589–608. doi:10.1037/0033-295X.100.4.589
- Comaskey, E. M., Savage, R. S., & Abrami, P. C. (2009). A randomised efficacy study of Web-based synthetic and analytic programmes among disadvantaged urban kindergarten children. *Journal of Research in Reading*, 32, 92–108. doi:10.1111/j.1467-9817.2008.01383.x
- Cuban, L. (2001). *Oversold and underused: Computers in the classroom*. Cambridge, MA: Harvard University Press.
- Davidson, M. R., Fields, M. K., & Yang, J. (2009). A randomized trial study of a preschool literacy curriculum: The importance of implementation. *Journal of Research on Educational Effectiveness*, 2, 177–208. doi:10.1080/19345740902770028
- Deault, L. (2011). *Effective classroom contexts to develop literacy and attention skills for typical and at-risk first grade students* (Unpublished doctoral dissertation). McGill University, Montreal, Canada.
- Deault, L., & Savage, R. S. (2012). *Effective classroom contexts to develop literacy and attention skills for typical and at-risk first grade students*. Manuscript submitted for publication.
- Deault, L., Savage, R. S., & Abrami, P. C. (2009). Inattention and response to the ABRACADABRA Web-based literacy intervention. *Journal of Research on Educational Effectiveness*, 2, 250–286. doi:10.1080/19345740902979371
- Dede, C. (1996). Emerging technologies and distributed learning. *American Journal of Distance Education*, 10, 4–36. doi:10.1080/08923649609526919
- Di Stasio, M. R., Savage, R. S., & Abrami, P. C. (2012). A follow up study of the ABRACADABRA web based literacy intervention in Grade 1. *Journal of Research in Reading*, 35, 169–86. doi:10.1111/j.1467-9817.2010.01469.x
- Dynarski, M., Agodini, R., Heavyside, S., Novak, T., Carey, N., & Campuzano, L., . . . Sussex, W. (2007). *Effectiveness of reading and mathematics software products: Findings from the first student cohort*. Washington, DC: U. S. Department of Education, Institute of Education Sciences.
- Ecalfe, J., Magnan, A., & Calmus, C. (2009). Lasting effects on literacy skills with a computer-assisted learning using syllabic units in low-progress readers. *Computers and Education*, 52, 554–561. doi:10.1016/j.compedu.2008.10.010
- Ehri, L. C. (2005). Learning to read words: Theory, findings, and issues. *Scientific Studies of Reading*, 9, 167–188. doi:10.1207/s1532799xssr0902_4

- Ehri, L., Nunes, S., Willows, D., Schuster, B. V., Yaghoub-Zadeh, Z., & Shanahan, T. (2001). Phonemic awareness instruction helps children learn to read: Evidence from the National Reading Panel's meta-analysis. *Reading Research Quarterly*, 36, 250–287. Retrieved from <http://www.jstor.org/stable/748111>. doi:10.1598/RRQ.36.3.2
- Fry, E. B., Kress, J. E., & Fountoukidis, D. L. (2000). *The reading teacher's book of lists* (4th ed.). Paramus, NJ: Prentice-Hall.
- Fugate, M. H. (2003). Review of the group reading assessment and diagnostic evaluation. In B. S. Plak, J. C. Impara, & R. A. Spies (Eds.), *The fifteenth mental measurements yearbook* (pp. 425–429). Lincoln, NE: Buros Institute of Mental Measurements.
- Fullan, M. (2001). *The new meaning of educational change* (3rd ed.). New York, NY: Teachers College Press.
- Grant, A., Wood, E., Gottardo, A., Evans, M. E., Phillips, L., & Savage, R. S. (2012). Assessing the content and quality of commercially available reading software programs: Do they have the fundamental structure to promote the development of early reading skills in children? *NHSA Dialog*, 15(4), 319–342. doi:10.1080/15240754.2012.725487
- Harasim, L., Hiltz, S. R., Teles, L., & Turoff, M. (1995). *Learning networks: A field guide to teaching and learning online*. Cambridge, MA: Massachusetts Institute of Technology Press.
- Hipps, G., Abrami, P. C., Savage, R. S., Cerna, N., & Jorgensen, A. (2005). ABRACADABRA: The research, design and development of Web-based early literacy software (pp. 89–112). In S. Pierre (Ed.), *DIVA: Innovations et tendances en technologies de formation et d'apprentissage* [DIVA: Innovations and trends in education and training technologies]. Montreal, Quebec, Canada: Presses Internationales Polytechnique.
- Hox, J. J. (2010). *Multilevel analysis: Techniques and applications* (2nd ed.). New York, NY: Routledge.
- Hulme, C., Snowling, M., Caravolas, M., & Carroll, J. (2005). Phonological skills are (probably) one cause of success in learning to read: A comment on Castles and Coltheart. *Scientific Studies of Reading*, 9, 351–365. doi:10.1207/s1532799xssr0904_2
- Jamieson, D. G. (2006). Literacy in Canada. *Paediatrics & Child Health*, 11, 573–574.
- Kaminski, R. A., & Good, R. H. (1996). Toward a technology for assessing basic early literacy skills. *School Psychology Review*, 25, 215–227.
- Kay, R. H. (2008). Exploring the relationship between emotions and the acquisition of computer knowledge. *Computers & Education*, 50, 1269–1283. doi:10.1016/j.compedu.2006.12.002
- Lane, K. L., Bocian, K. M., MacMillan, D. L., & Gresham, F. M. (2004). Treatment integrity: An essential—but often forgotten—component of school-based interventions. *Preventing School Failure*, 48, 36–43. doi:10.3200/PSFL.48.3.36-43
- Lewis, R. B., Ashton, T. M., Haapa, B., Kieley, C. L., & Fielden, C. (1999). Improving the writing skills of students with learning disabilities: Are word processors with spelling and grammar checkers useful? *Learning Disabilities: A Multidisciplinary Journal*, 9, 87–98.
- MacArthur, C. A., Ferretti, R. P., Okolo, C. M., & Cavalier, A. R. (2001). Technology applications for students with literacy problems: A critical review. *The Elementary School Journal*, 101, 273–301.
- Mayer, R. E. (2001). *Multimedia learning*. New York, NY: Cambridge Press. doi:10.1017/CBO9781139164603
- McBride, J. R., Ysseldyke, J., Milone, M., & Stickney, E. (2010). Technical adequacy and cost benefit of four measures of early literacy. *Canadian Journal of School Psychology*, 25, 189–204. doi:10.1177/0829573510363796
- Miller, G. A. (1969). Psychology as a means of promoting human welfare. *American Psychologist*, 24, 1063–1075. doi:10.1037/h0028988
- Mowbray, C. T., Holter, M. C., Teague, G. B., & Bybee, D. (2003). Fidelity criteria: Development, measurement, and validation. *The American Journal of Evaluation*, 24, 315–340. doi:10.1177/109821400302400303
- Mueller, J., Wood, E., & Willoughby, T. (2008). The integration of computer technology in the classroom. In T. Willoughby & E. Wood (Eds.), *Children's learning in a digital world* (pp. 272–297). Oxford, England: Blackwell. doi:10.1002/9780470696682.ch11
- Mueller, J., Wood, E., Willoughby, T., Ross, C., & Specht, J. (2008). Identifying discriminating variables between teachers who fully integrate computers and teachers with limited integration. *Computers & Education*, 51, 1523–1537. doi:10.1016/j.compedu.2008.02.003
- National Reading Panel. (2000). Teaching children to read: An evidence-based assessment of the scientific research literature on reading and its implications for reading instruction: Reports of the subgroups (National Institute of Health Publication No. 00-4769). Washington, DC: National Institute of Child Health and Human Development.
- Perfetti, C. (2007). Reading ability: Lexical quality to comprehension. *Scientific Studies of Reading*, 11, 357–383. doi:10.1080/10888430701530730
- Pressley, M. (1998). *Reading instruction that works*. New York, NY: Guilford Press.
- Pressley, M. (2002). Metacognition and self-regulated comprehension. In A. Farstrup & J. Samuels (Eds.), *What research says about reading instruction* (3rd ed., pp. 291–309). Newark, DE: International Reading Association.
- Puffer, S., Torgerson, D. J., & Watson, J. (2005). Cluster randomized controlled trials. *Journal of Evaluation in Clinical Practice*, 11, 479–483. doi:10.1111/j.1365-2753.2005.00568.x
- Rabiner, D. L., & Malone, P. S. (2004). The impact of tutoring on early reading achievement for children with and without attention problems. *Journal of Abnormal Child Psychology*, 32, 273–284. doi:10.1023/B:JACP.0000026141.20174.17
- Raudenbush, S. W., & Bryk, A. S. (2002). *Hierarchical linear models: Applications and data analysis methods*. Thousand Oaks, CA: Sage.
- Sandholtz, J. H., Ringstaff, C., & Dwyer, D. C. (1997). *Teaching with technology: Creating student-centered classrooms*. New York, NY: Teachers College Press.
- Sang, G., Valcke, M., van Braak, J., & Tondeur, J. (2010). Student teachers' thinking processes and ICT integration: Predictors of prospective teaching behaviors with educational technology. *Computers & Education*, 54, 103–112. doi:10.1016/j.compedu.2009.07.010
- Savage, R. S. (2012). Evidence-based reading interventions: Implementation issues for the 21st century. In B. Kelly & D. F. Perkins (Eds.), *The Cambridge handbook of implementation science for psychology in education* (pp. 277–297). Cambridge, England: Cambridge University Press.
- Savage, R. S., Abrami, P. C., Hipps, G., & Deault, L. (2009). A randomized controlled trial study of the ABRACADABRA reading intervention program in Grade 1. *Journal of Educational Psychology*, 101, 590–604. doi:10.1037/a0014700
- Savage, R. S., Carless, S., & Ferraro, V. (2007). Predicting curriculum and test performance at age 11 years from pupil background, baseline skills, and phonological awareness at age 5 years. *Journal of Child Psychology and Psychiatry and Allied Disciplines*, 48, 732–739. doi:10.1111/j.1469-7610.2007.01746.x
- Savage, R. S., Deault, L., Daki, J., & Aouad, J. (2011). Orthographic analogies and early reading: Evidence from a multiple clue word paradigm. *Journal of Educational Psychology*, 103, 190–205. doi:10.1037/a0021621
- Savage, R. S., Erten, O., Abrami, P. C., Hipps, G., Comaskey, E., & van Lierop, D. (2010). ABRACADABRA in the hands of teachers: The effectiveness of a Web-based literacy intervention in Grade 1 language arts programs. *Computers & Education*, 55, 911–922. doi:10.1016/j.compedu.2010.04.002
- Savage, R. S., & Stuart, M. (2006). A developmental model of reading acquisition based upon early scaffolding errors and subsequent vowel

- inferences. *Educational Psychology*, 26, 33–53. doi:10.1080/01443410500340983
- Scardamalia, M., & Bereiter, C. (1996). Adaptation and understanding: A case for new cultures of schooling. In S. Vosniadou, E. DeCorte, R. Glaser, & H. Mandl (Eds.), *International perspectives on the design of technology-supported learning environments* (pp. 149–163). Mahwah, NJ: Erlbaum.
- Seidenberg, M. S., & McClelland, J. L. (1989). A distributed, developmental model of word recognition and naming. *Psychological Review*, 96, 523–568. doi:10.1037/0033-295X.96.4.523
- Seymour, P. H. K. (1997). Foundations of orthographic development. In C. Perfetti, L. Rieben, & M. Fayol (Eds.), *Learning to spell: Research, theory, and practice across languages* (pp. 319–337). London, England: Erlbaum.
- Share, D. L. (1995). Phonological recoding and self-teaching: Sine qua non of reading acquisition. *Cognition*, 55, 151–218. doi:10.1016/0010-0277(94)00645-2
- Share, D. L. (2008). Orthographic learning, phonological recoding, and self-teaching. *Advances in Child Development and Behavior*, 36, 31–82. doi:10.1016/S0065-2407(08)00002-5
- Slavin, R. E., Cheung, A., Groff, C., & Lake, C. (2008). Effective reading programs for middle and high schools: A best-evidence synthesis. *Reading Research Quarterly*, 43, 290–322. doi:10.1598/RRQ.43.3.4
- Slavin, R. E., Lake, C., Chambers, B., Cheung, A., & Davis, S. (2009). Effective reading programs for the elementary grades: A best-evidence synthesis. *Review of Educational Research*, 79, 1391–1466. doi:10.3102/0034654309341374
- Smith, M. W., Dickinson, D. K., & Sangeorge, A. (2002). *The early language and literacy classroom observation*. Baltimore, MD: Brookes.
- Solso, R. L. (2001). *Cognitive psychology*. Boston, MA: Allyn & Bacon.
- Statistics Canada. (2006). 2006 census of population. Retrieved from <http://www12.statcan.gc.ca/census-recensement/2006/index-eng.cfm>
- Stein, M. L., Berends, M., Fuchs, D., McMaster, K., Sáenz, L., Yen, L., . . . Compton, D. L. (2008). Scaling up an early reading program: Relationships among teacher support, fidelity of implementation, and student performance across different sites and years. *Educational Evaluation and Policy Analysis*, 30, 368–388. doi:10.3102/0162373708322738
- Tabachnick, B. G., & Fidell, L. S. (2007). *Using multivariate statistics* (5th ed.). Boston, MA: Allyn & Bacon.
- Torgerson, C. J. (2007). The quality of systematic reviews of effectiveness in literacy learning in English: A “tertiary” review. *Journal of Research in Reading*, 30, 287–315. doi:10.1111/j.1467-9817.2006.00318.x
- Torgerson, C. J., Brooks, G., & Hall, J. (2006). *A systematic review of the research literature on the use of phonics in the teaching of reading and spelling*. London, England: Department for Education and Skills. Retrieved from http://www.dfes.gov.uk/research/data/uploadfiles/RR711_.pdf
- Torgerson, C. J., & Zhu, D. (2003). A systematic review and meta-analysis of the effectiveness of ICT on literacy learning in English, 5–16. In Research Evidence in Education Library. London, England: EPPI-Centre, Social Science Research Unit, Institute of Education.
- Wagner, R. K., Torgesen, J. K., & Rashotte, C. A. (1999). *The Comprehensive Test of Phonological Processing (CTOPP): Examiner’s manual*. Austin, TX: Pro-Ed.
- Waks, L. J. (2007). The concept of fundamental educational change. *Educational Theory*, 57, 277–295. doi:10.1111/j.1741-5446.2007.00257.x
- Waterman, B. B. (2003). Review of the group reading assessment and diagnostic evaluation. In B. S. Plak, J. C. Impara, & R. A. Spies (Eds.), *The fifteenth mental measurements yearbook* (pp. 239–241). Lincoln, NE: Buros Institute of Mental Measurements.
- Williams, K. T. (2001). *Group reading assessment and diagnostic evaluation: Teacher’s scoring & interpretive manual*. Circle Pines, MN: American Guidance Service.
- Wolgemuth, J. R., Helmer, J., Bottrell, C., Harper, H., & Lea, T. (2012). *A multi-site randomised control trial to examine the impact of ABRACA-DABRA on indigenous literacy in Australia*. Manuscript submitted for publication.
- Wolgemuth, J. R., Savage, R. S., Helmer, J., Bottrell, C., Emmett, S., Lea, T., . . . Abrami, P. (2011). Using computer-based instruction to improve indigenous early literacy in Northern Australia: A quasi-experimental study. *Australasian Journal of Educational Technology*, 27, 727–750.
- Wood, E., Specht, J., Willoughby, T., & Mueller, J. (2008). Integrating computer technology in early childhood education environments: Issues raised by early childhood educators. *Alberta Journal of Educational Research*, 54, 210–226.
- Wozney, L., Venkatesh, V., & Abrami, P. (2006). Implementing computer technologies: Teachers’ perceptions and practices. *Journal of Technology and Teacher Education*, 14, 173–207. Retrieved from <http://doe.concordia.ca/csllp/wozneyetaljte141.pdf>

Received August 5, 2011

Revision received September 17, 2012

Accepted October 1, 2012 ■